MATE6301
Nanomaterials

Week 5: Magnetism – Origin, Types and Fundamentals of Nano-magnetism

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Learning Objective

After successfully completing this module you should be able to:

FABRICATE/CHARACTERISE MAGNETIC and PLASMONIC NANOCOMPOSITES

- LO4.

Design, fabricate, and characterise functional nanomaterials, including porous gold nanostructures and magnetic nanocomposites, and analyse their optical and magnetic properties for practical applications

- LO5.

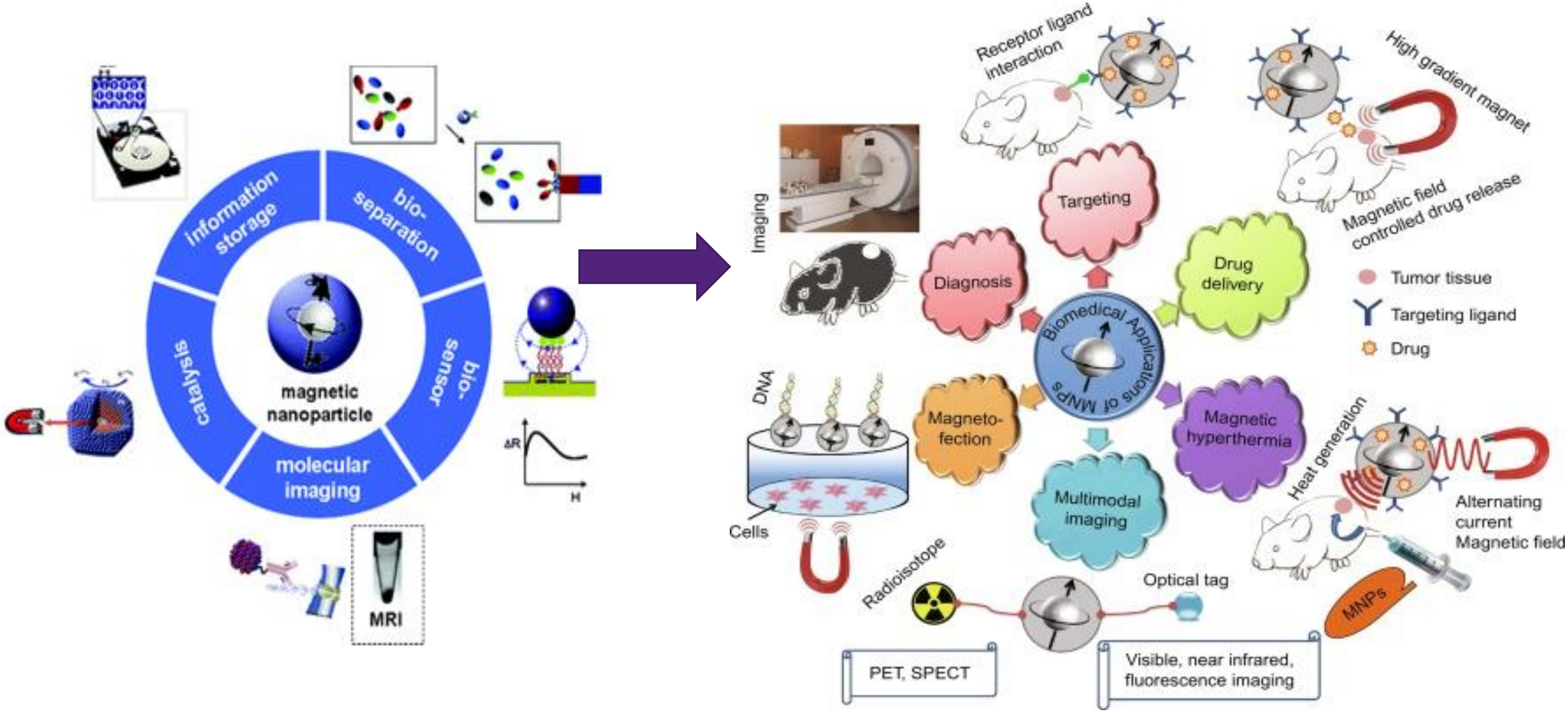
Select appropriate magnetic and optical nanostructures to meet the requirements of designed applications.

L-6 - Outline

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- Origin and fundamentals of magnetism
- Magnetic materials and their types
- Magnetic hysteresis
- Bulk (macro) magnetism (Ferromagnetism) and Nano magnetism (superparamagnetism)
- Synthesis of Magnetic Nanoparticles.
- Parameters influence magnetic properties of nanoparticles/nanocomposites
- Effect of annealing temperatures on magnetic phase transition and magnetic properties of nanoparticles
- Information about the Lab practical.

Why we need to learn Nanomagnetism

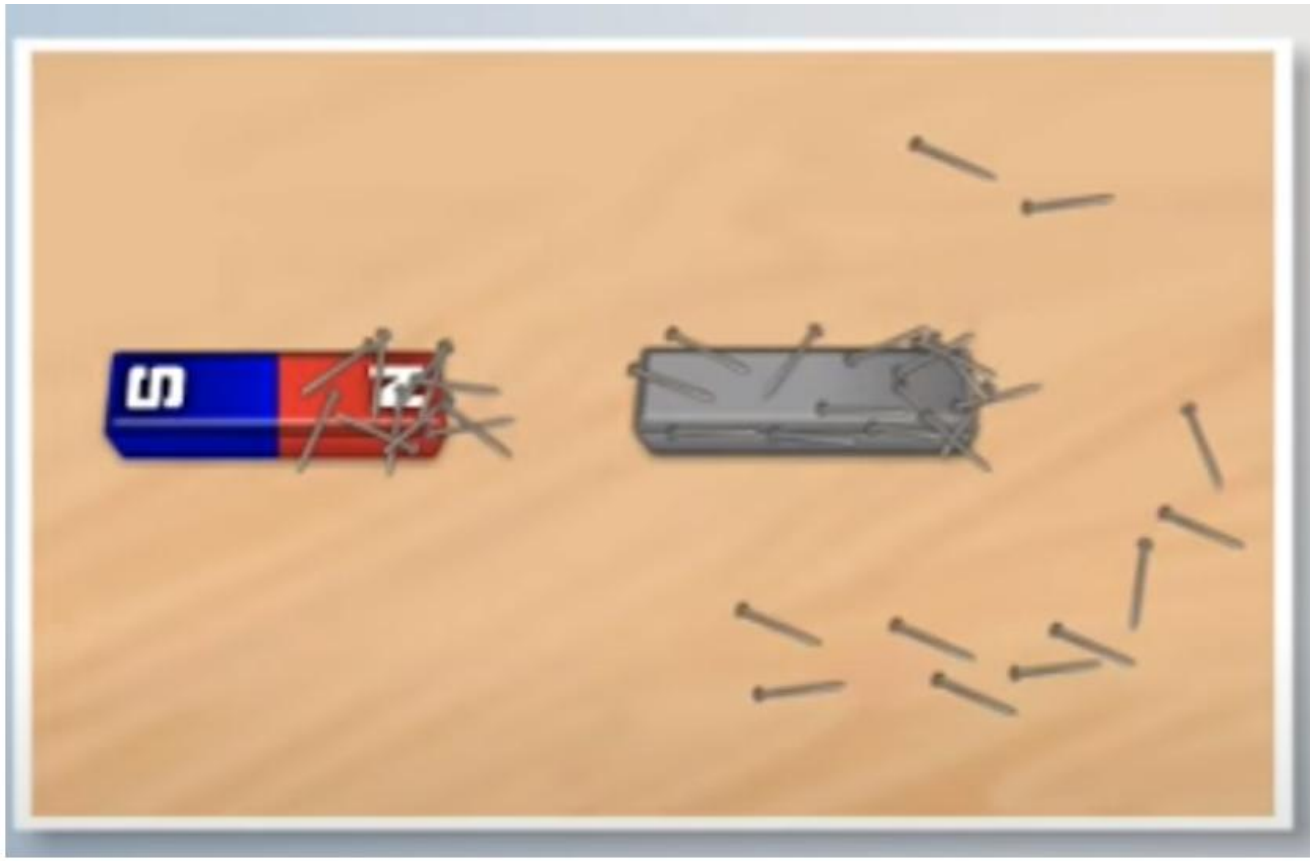


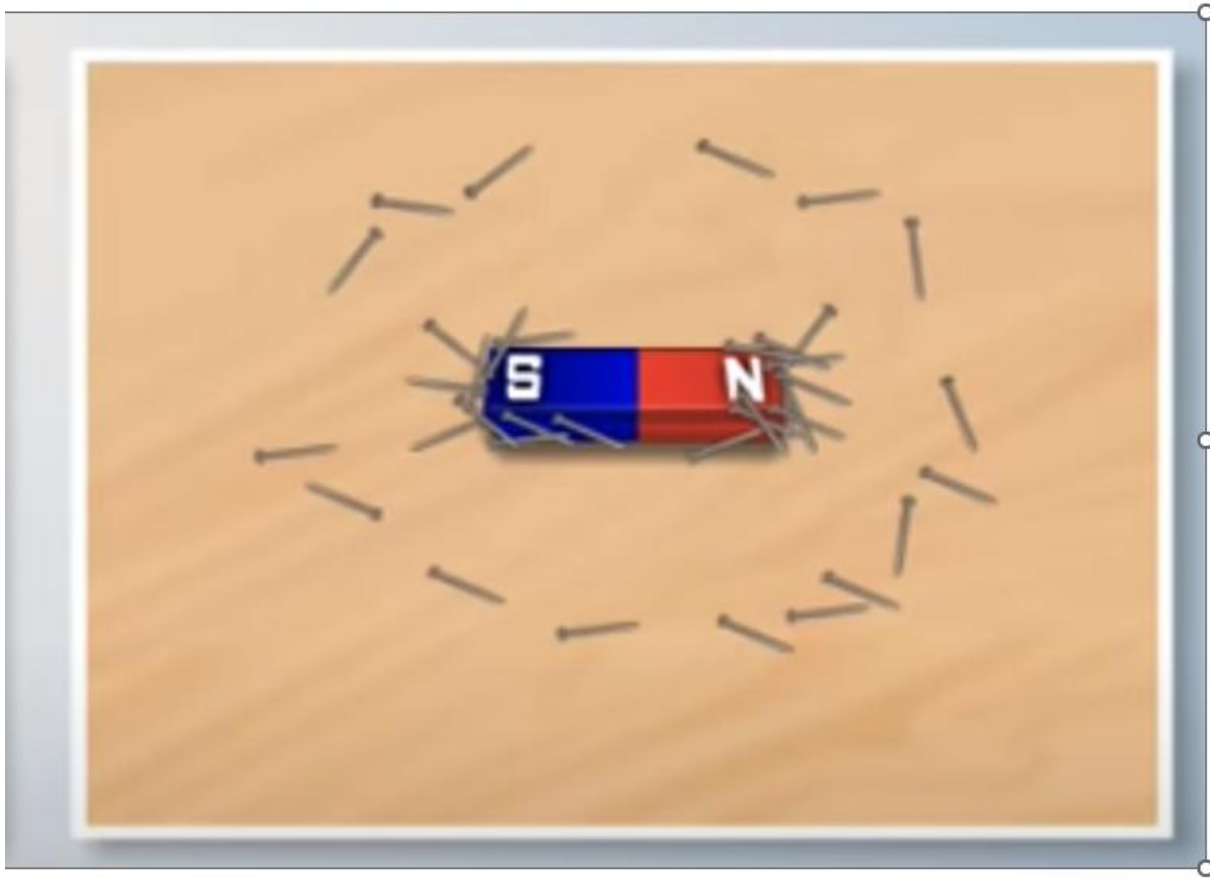
https://www.cd-bioparticles.com/t/Properties-and-Applications-of-Magnetic-Nanoparticles_55.html

Understanding the facts of Magnetism

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Why such behaviour?

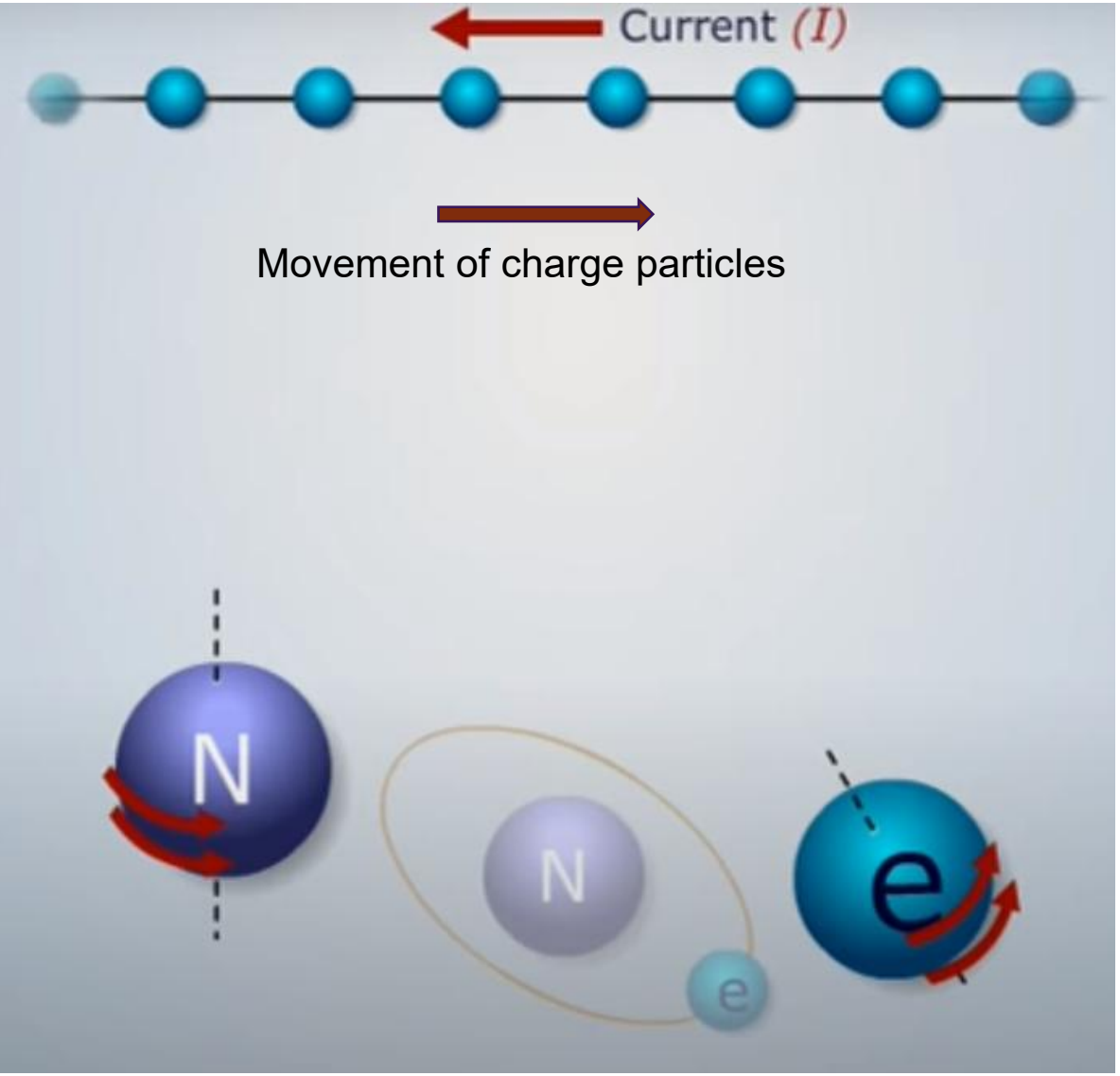


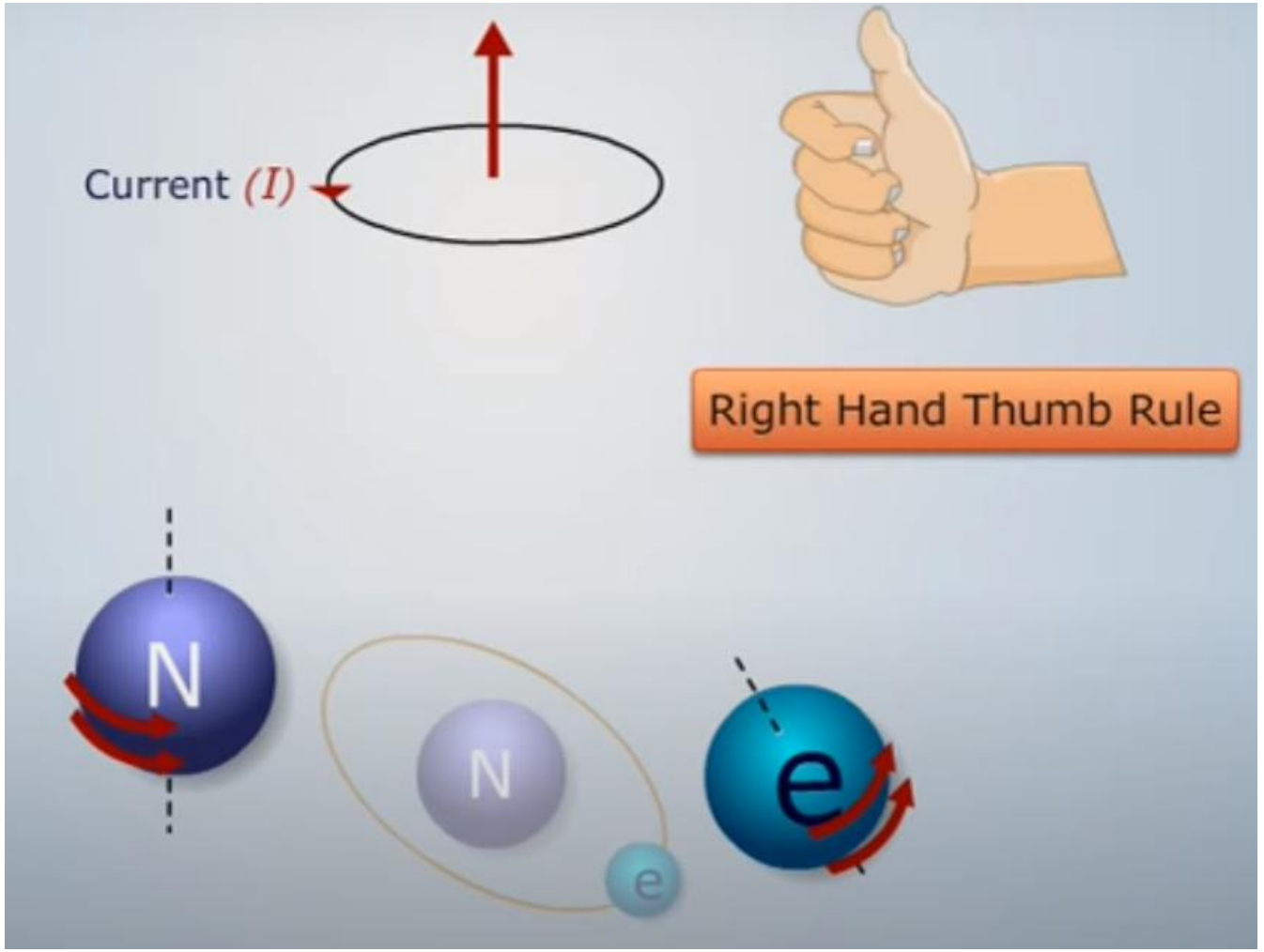


Understanding the facts of Magnetism

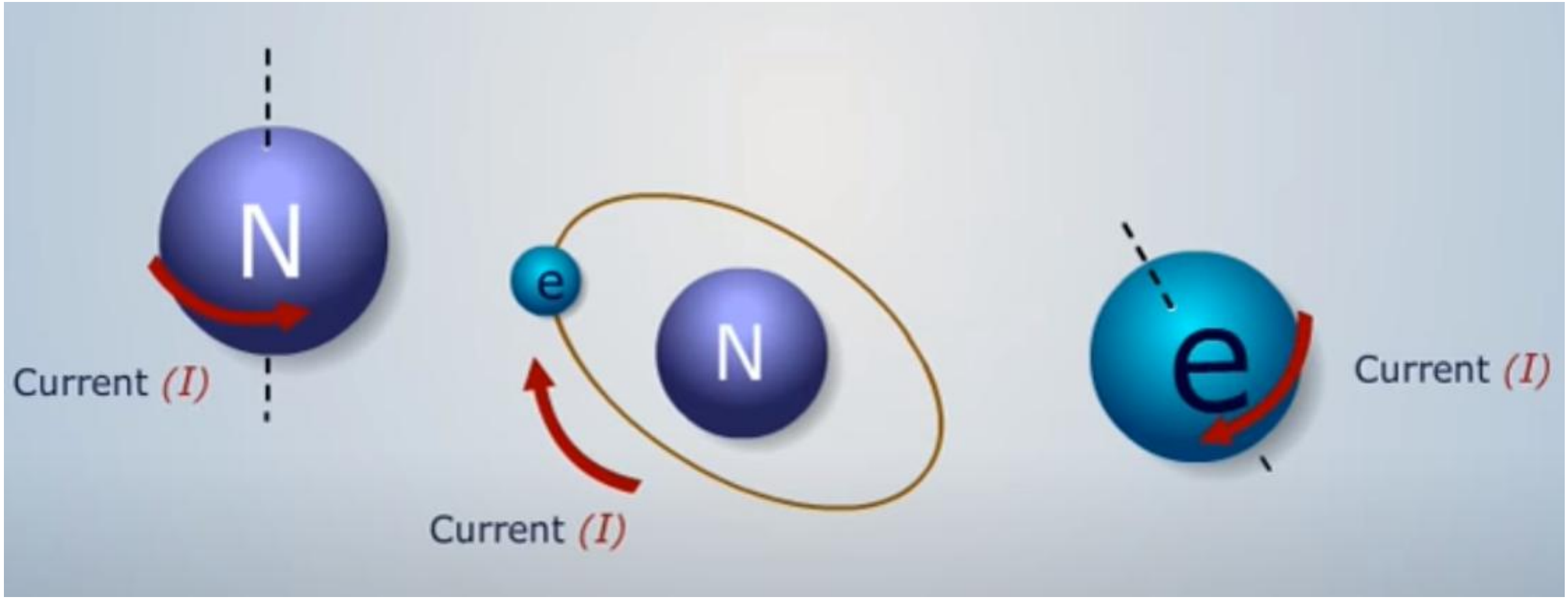
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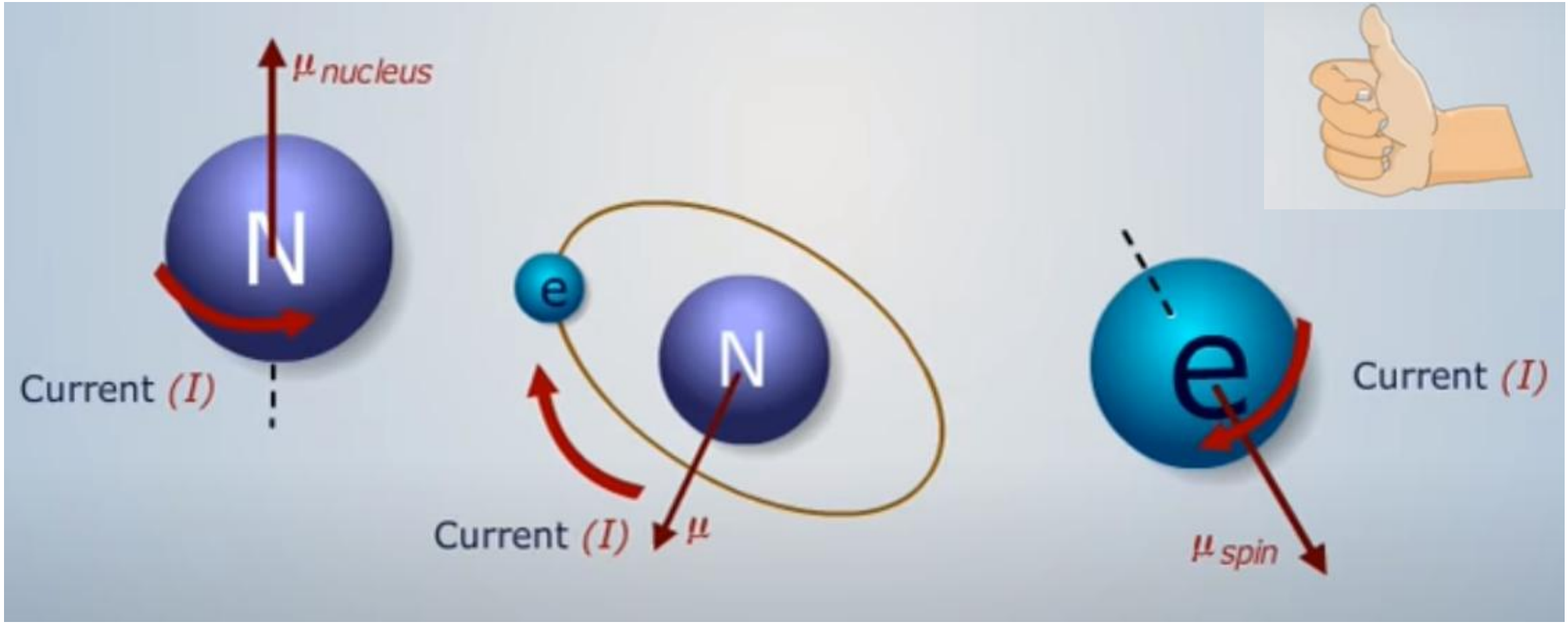




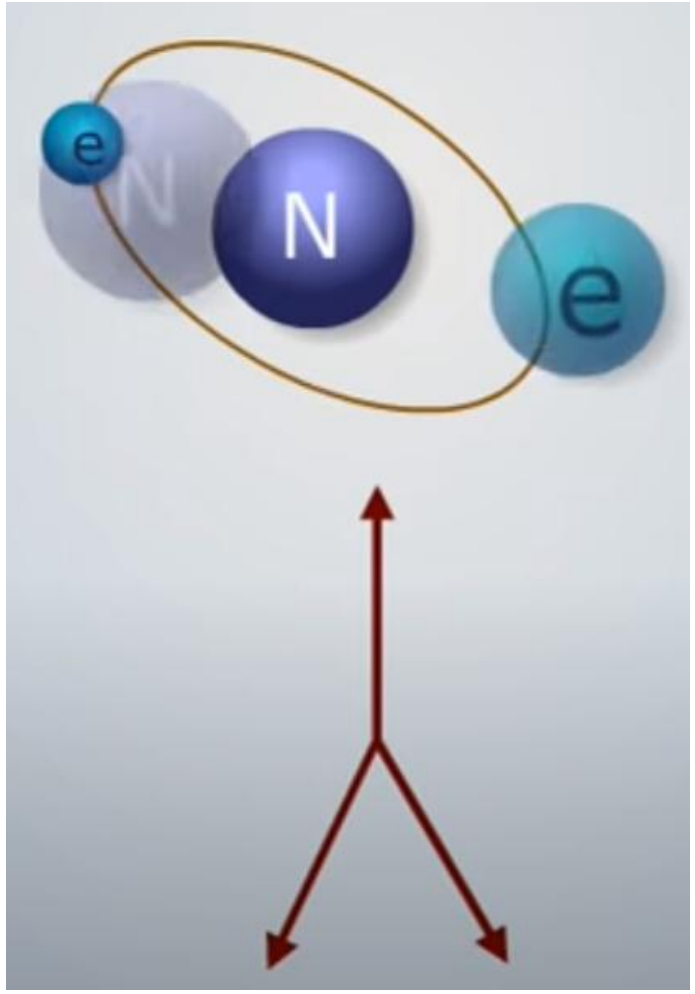
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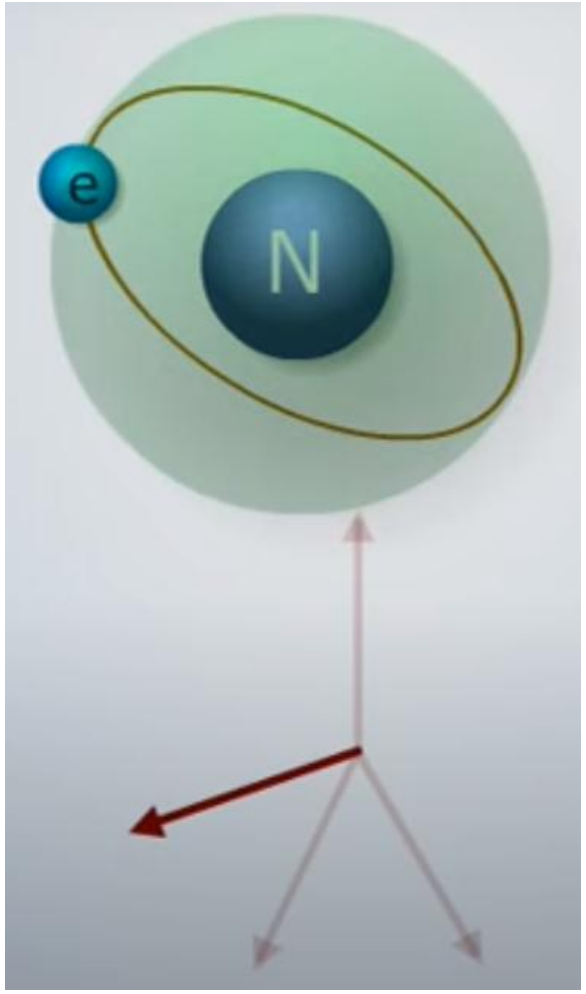


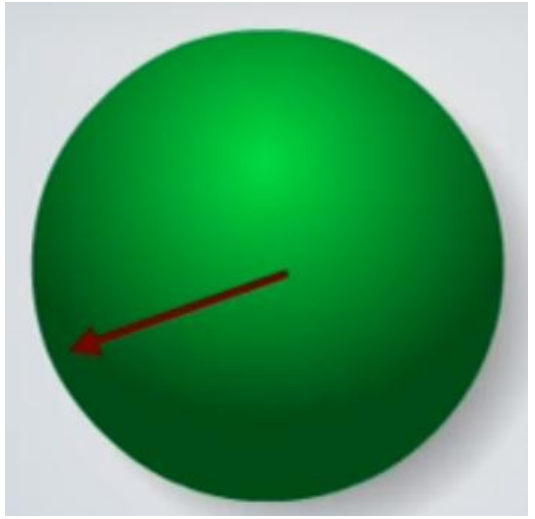
Why such behaviour?



Why such behaviour?

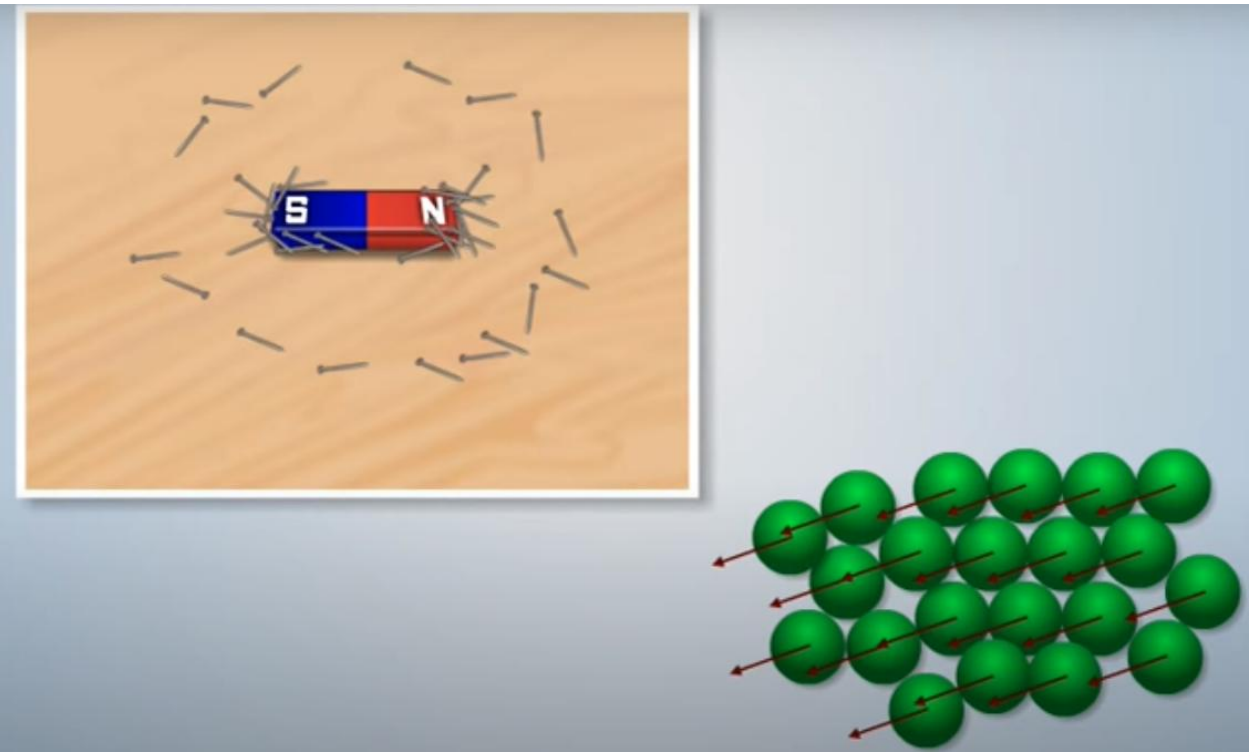


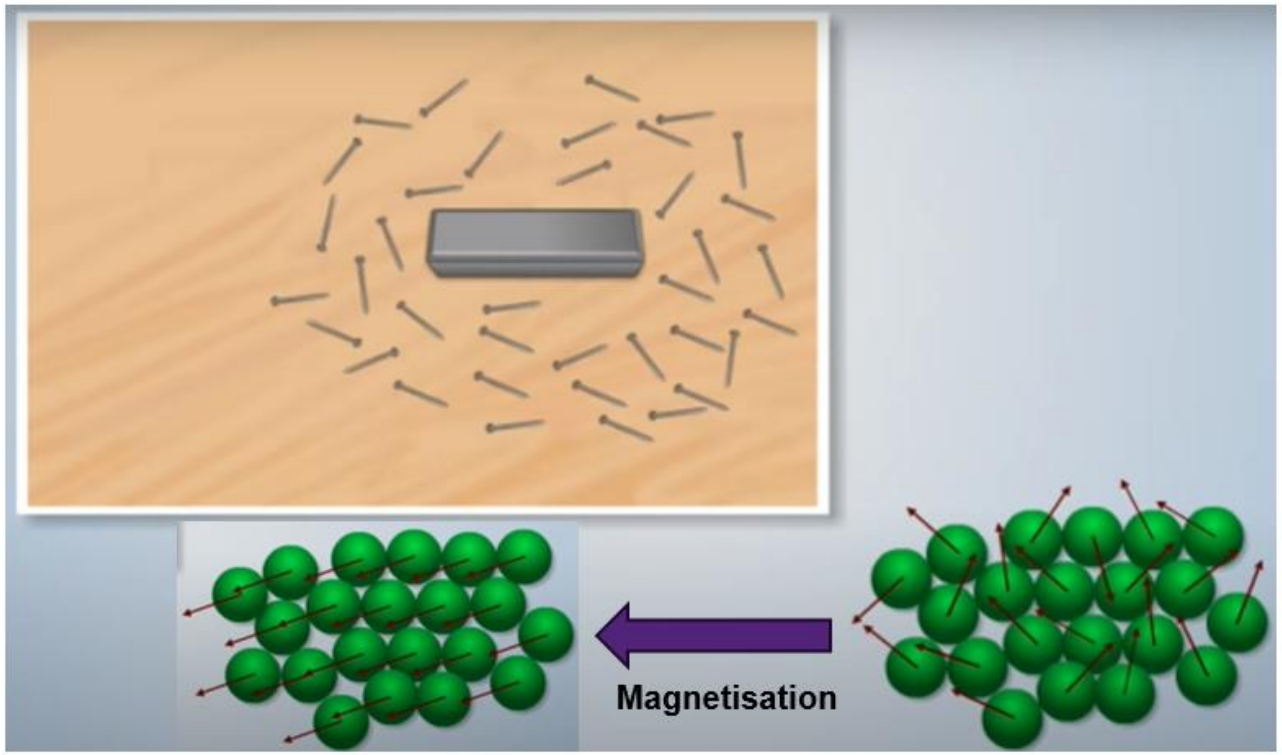




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Why such behaviour?





Understanding the facts of Magnetism

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Why such behaviour?

Mathematically

Magnetic Moment due to Revolving Electrons

$$\mu = -\frac{ehl}{4\pi mc}$$

Magnetic Moment due to Spinning Electrons:

$$\mu_{spin} = \frac{eh}{4\pi mc}$$

Magnetic Moment due to Spinning Nucleus:

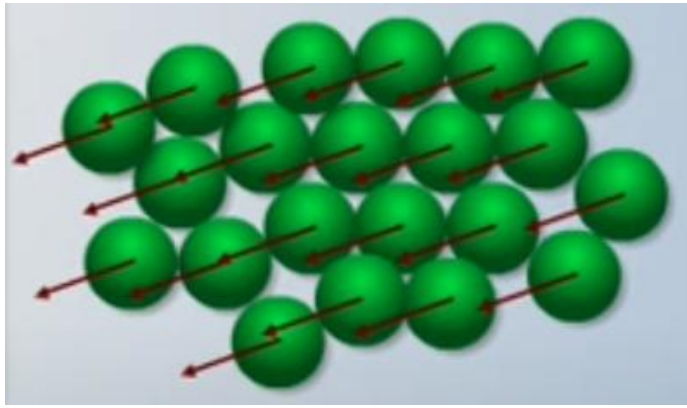
$$\mu_{nucleus} = \frac{eh}{4\pi m_p c}$$

Where,

e = Charge on an Electron
h = Planck's Constant
l = Orbital Quantum Number(= 0,1,2,....)
m = Mass of an Electron
c = Velocity of Light (= 3 X 10^8 m/s)
m_p = Mass of Proton

Keywords

- *Magnetic Moments*
- *Spinning and Revolving Electrons*
- *Spinning Nucleus*



ATOMIC ORIGIN OF MAGNETISM in any **Magnetic Material** can be explained by taking into consideration the *Magnetic Moments* of the *Spinning* and *Revolving Electrons* and also that of the *Spinning Nucleus.*

Basic Magnetism

We are all familiar with some basic properties of electron and atoms

Magnetic dipole of electrons → macroscopic behavior of matter

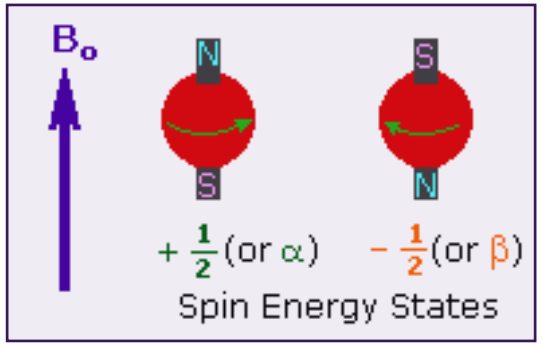
Atom → Electron —Spin→ Magnetic moment → Magnetism

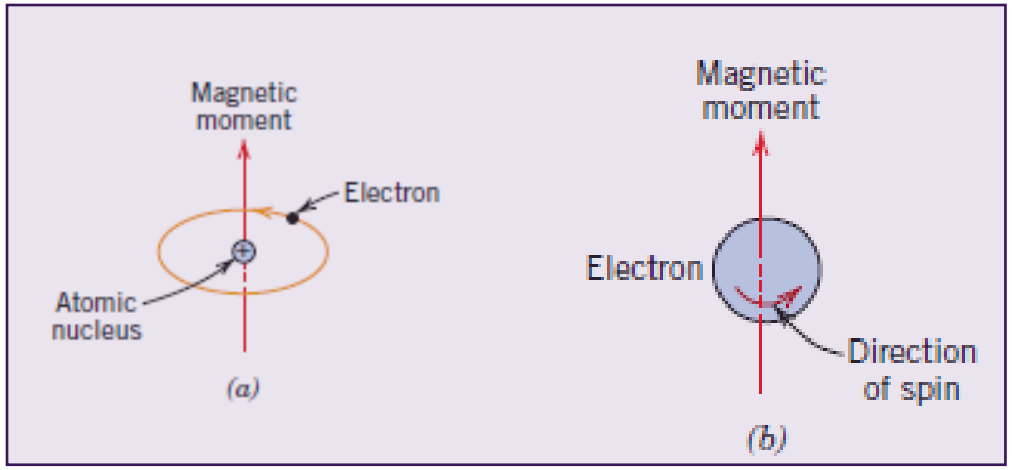
Each electron in an atom may be thought of as being a small magnet having permanent orbital and spin magnetic moments

Net atomic magnetic moment = some of moments from all electrons

Orbital motion around the nucleus; being a moving charge, an electron may be considered to be a small current loop, generating a very small magnetic field having a magnetic moment along its axis of rotation.

- Nucleus at the center, electrons rotating on orbits
- Electron is negatively charged
- Electron has an intrinsic angular momentum (spin)





Reference: Callister 8th Ed.; G. Sciolla – MIT

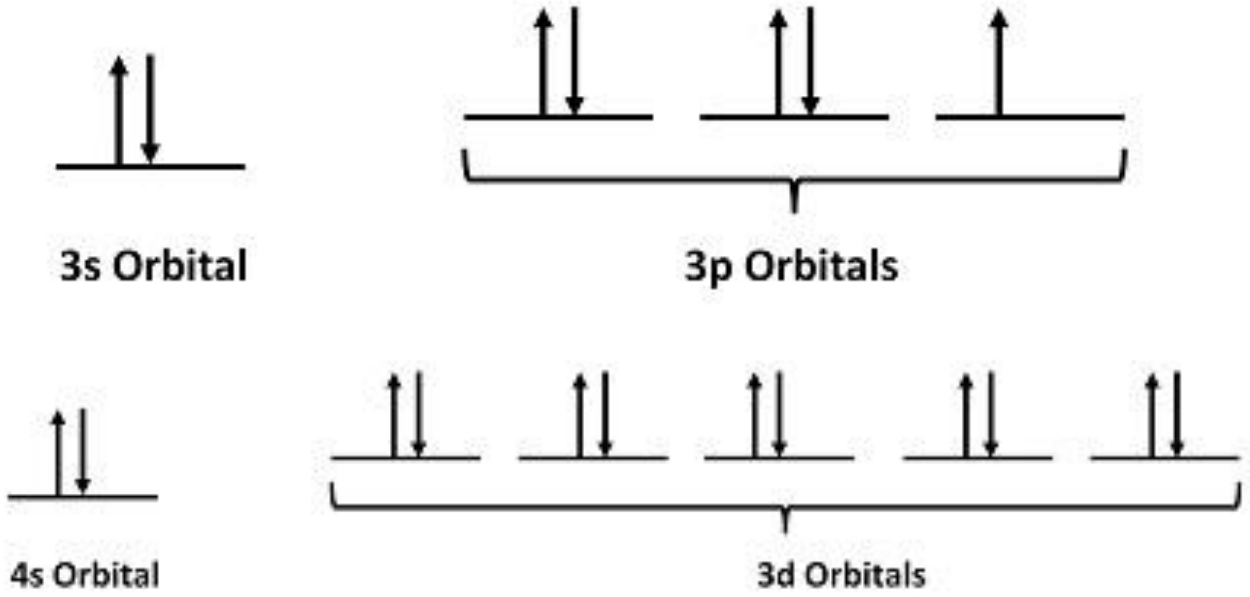
A magnetic moment is a vector quantity, with a magnitude and a direction.

Origins of Magnetism: From Atomic Model

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Every materials has orbital and spinning electron, in that case, all materials should be magnetic, right?

Of course : No!
Why?



The arrangement and interaction of these electrons differ significantly.

Paramagnetic

[↑] [↓] [↑] [↓] [↑] ← unpaired electrons, random directions
Random alignment → weak response to magnetic field

Ferromagnetic

[↑] [↑] [↑] [↑] ← unpaired electrons, spins aligned
Spontaneous alignment → strong permanent magnetization

- Number of unpaired electrons alone doesn't guarantee ferromagnetism.
- It's the interaction between spins (exchange coupling) that creates ferromagnetism.
- Net magnetization is important for the consideration of higher or lower magnetisation

Exchange energy interactions

- Exchange energy is the energy associated with the alignment of spins between neighboring atoms.
- It determines whether the material will exhibit strong magnetism or not.
- The exchange interactions between neighboring atoms are strong and lead to a parallel alignment of spins over large regions, resulting in strong magnetism.

↑ ↑ ↑ ↑ ↑ ↑ ↑ ↑ ↑ ↑	↑ ↓ ↑ ↓ ↑ ↓ ↑ ↓ ↑ ↓ ↑ ↓
All arrows aligned Strong magnetism	Arrows cancel out Weak or no magnetism

- Despite having more unpaired electrons, the exchange interactions are not not strong. The spins in atoms tend to align in an antiferromagnetic arrangement (neighboring spins point in opposite directions), which cancels out the overall magnetism.

Origins of Magnetism: From Atomic Model

Martin S. Silberberg, *Chemistry: The Molecular Nature of Matter and Change*, 2nd Edition. Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

Table 8.4 Partial Orbital Diagrams and Electron Configurations*

Atomic Number	Element	Partial Orbital Diagram (4s, 3d, and 4p Sublevels Only) 4s	3d	4p	Full Electron Configuration	Condensed Electron Configuration
19	K	↑			$[1s^2 2s^2 2p^6 3s^2 3p^6]\ 4s^1$	[Ar] $4s^1$
20	Ca	↑↓			$[1s^2 2s^2 2p^6 3s^2 3p^6]\ 4s^2$	[Ar] $4s^2$
21	Sc	↑↓	↑		$[1s^2 2s^2 2p^6 3s^2 3p^6]\ 4s^2 3d^1$	[Ar] $4s^2\ 3d^1$
22	Ti	↑↓	↑ ↑		$[1s^2 2s^2 2p^6 3s^2 3p^6]\ 4s^2 3d^2$	[Ar] $4s^2\ 3d^2$
23	V	↑↓	↑ ↑ ↑		$[1s^2 2s^2 2p^6 3s^2 3p^6]\ 4s^2 3d^3$	[Ar] $4s^2\ 3d^3$
24	Cr	↑	↑ ↑ ↑ ↑ ↑		$[1s^2 2s^2 2p^6 3s^2 3p^6]\ 4s^1 3d^5$	[Ar] $4s^1\ 3d^5$
25	Mn	↑↓	↑ ↑ ↑ ↑ ↑		$[1s^2 2s^2 2p^6 3s^2 3p^6]\ 4s^2 3d^5$	[Ar] $4s^2 3d^5$
26	Fe	↑↓	↑↓ ↑ ↑ ↑ ↑		$[1s^2 2s^2 2p^6 3s^2 3p^6]\ 4s^2 3d^6$	[Ar] $4s^2 3d^6$
27	Co	↑↓	↑↓ ↑↓ ↑ ↑ ↑		$[1s^2 2s^2 2p^6 3s^2 3p^6]\ 4s^2 3d^7$	[Ar] $4s^2 3d^7$
28	Ni	↑↓	↑↓ ↑↓ ↑↓ ↑ ↑		$[1s^2 2s^2 2p^6 3s^2 3p^6]\ 4s^2 3d^8$	[Ar] $4s^2 3d^8$
29	Cu	↑	↑↓ ↑↓ ↑↓ ↑↓ ↑↓		$[1s^2 2s^2 2p^6 3s^2 3p^6]\ 4s^1 3d^{10}$	[Ar] $4s^1\ 3d^{10}$
30	Zn	↑↓	↑↓ ↑↓ ↑↓ ↑↓ ↑↓		$[1s^2 2s^2 2p^6 3s^2 3p^6]\ 4s^2 3d^{10}$	[Ar] $4s^2 3d^{10}$
31	Ga	↑↓	↑↓ ↑↓ ↑↓ ↑↓ ↑↓	↑	$[1s^2 2s^2 2p^6 3s^2 3p^6]\ 4s^2 3d^{10} 4p^1$	[Ar] $4s^2 3d^{10} 4p^1$
32	Ge	↑↓	↑↓ ↑↓ ↑↓ ↑↓ ↑↓	↑ ↑	$[1s^2 2s^2 2p^6 3s^2 3p^6]\ 4s^2 3d^{10} 4p^2$	[Ar] $4s^2 3d^{10} 4p^2$
33	As	↑↓	↑↓ ↑↓ ↑↓ ↑↓ ↑↓	↑ ↑ ↑	$[1s^2 2s^2 2p^6 3s^2 3p^6]\ 4s^2 3d^{10} 4p^3$	[Ar] $4s^2 3d^{10} 4p^3$
34	Se	↑↓	↑↓ ↑↓ ↑↓ ↑↓ ↑↓	↑↓ ↑ ↑	$[1s^2 2s^2 2p^6 3s^2 3p^6]\ 4s^2 3d^{10} 4p^4$	[Ar] $4s^2 3d^{10} 4p^4$
35	Br	↑↓	↑↓ ↑↓ ↑↓ ↑↓ ↑↓	↑↓ ↑↓ ↑	$[1s^2 2s^2 2p^6 3s^2 3p^6]\ 4s^2 3d^{10} 4p^5$	[Ar] $4s^2 3d^{10} 4p^5$
36	Kr	↑↓	↑↓ ↑↓ ↑↓ ↑↓ ↑↓	↑↓ ↑↓ ↑↓	$[1s^2 2s^2 2p^6 3s^2 3p^6]\ 4s^2 3d^{10} 4p^6$	[Ar] $4s^2 3d^{10} 4p^6$

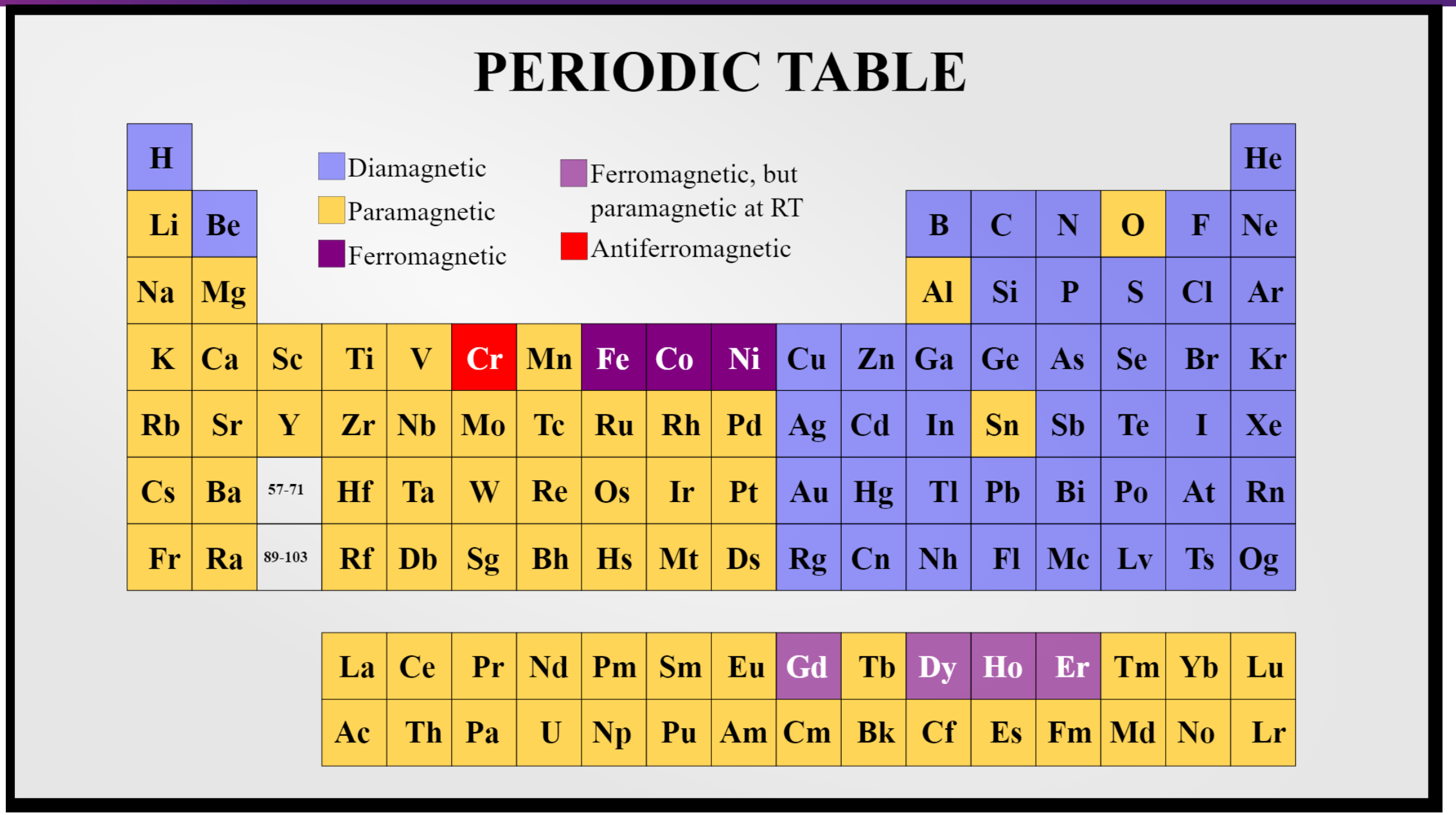
*Colored type indicates sublevel(s) whose occupancy changes when last electron is added.

In general, materials that have all paired electrons in the atoms and thus have no net magnetic moment are called **diamagnetic materials.** Diamagnetic materials include zinc, gold, mercury, and bismuth.

Substances with unpaired electrons are called **"paramagnetic"** as they are attracted towards a magnetic field. Most materials with one or more unpaired electrons are at least slightly magnetic. Materials with a small attraction to a magnet are called **paramagnetic materials**, and those with a strong attraction are called **ferromagnetic materials**. Aluminum, platinum, and manganese are some paramagnetic materials. Iron, cobalt, and nickel are examples of ferromagnetic materials.

Types of Magnetism

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http://periodic-table-1.blogspot.com/2014/12/periodic-table-magnetic-metals.html

Basic Terminology

The **externally applied magnetic field**, sometimes called the magnetic field strength**,** is designated by *H*. If the magnetic field is generated by means of a cylindrical coil (or solenoid) consisting of *N* closely spaced turns, having a length *l*, and carrying a current of magnitude *I*,:

$$H = \frac{NI}{l}$$

The units of *H* are ampere-turns per meter, or just amperes per meter. H is produced by an electric charge in motion by i) current flowing in a conductor ii) orbital movement and spin of electron

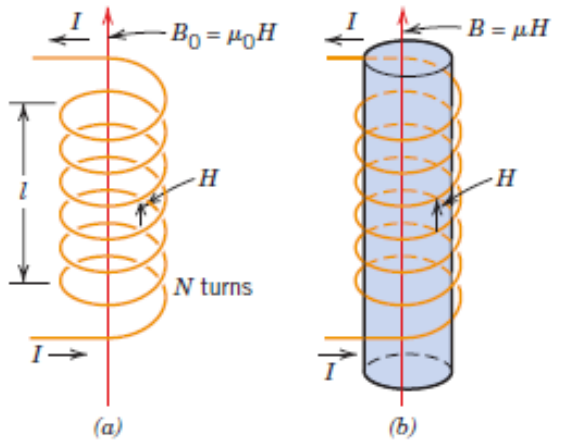
The **magnetic induction, or magnetic flux density,** denoted by *B*, represents the magnitude of the internal field strength within a substance that is subjected to an *H* field. The units for *B* are teslas [or webers per square meter (Wb/m^2)].

Both *B* and *H* are field vectors, being characterized not only by magnitude, but also by direction in space.

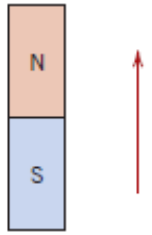
Magnetisation (M): process of being magnetized (iron gets magnetized under external magnetic field)

Magnetic susceptibility is a dimensionless proportionality constant that indicates the degree of magnetization of a material in response to an applied **magnetic** field. ... A closely related parameter is the **permeability**, which expresses the total magnetization of material and volume. (ratio of M/H)

Permeability **(ability to create internal magnetic field)** is the measure of the ability of a material to support the formation of a **magnetic** field within itself. Iron gets magnetized (B) if it is kept in external magnetic field (H). (Ratio of B/H)



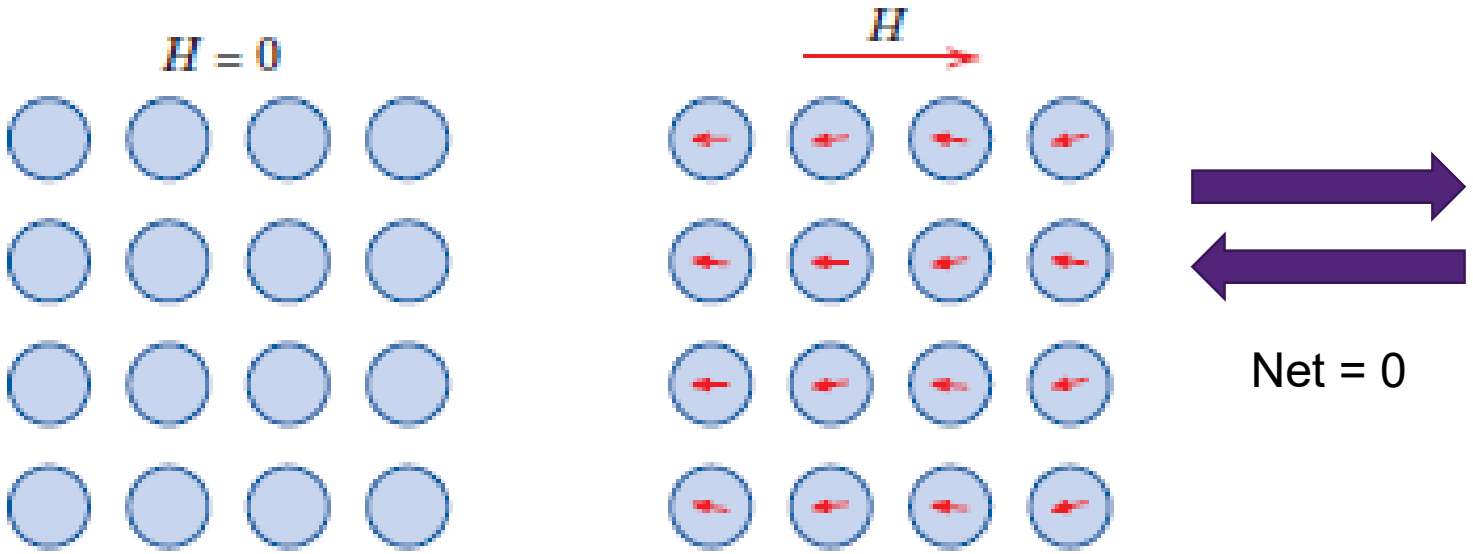
(a) The magnetic field *H* as generated by a cylindrical coil is dependent on the current *I*, the number of turns *N*, and the coil length *l*. The magnetic flux density B_0 in the presence of a vacuum is equal to $\mu_0 H$, where μ_0 is the permeability of a vacuum, $4\pi \times 10^{-7}$ H/m (*b*) The magnetic flux density *B* within a solid material is equal to μH, where μ is the permeability of the solid material.



The magnetic moment as designated by an arrow

Reference: Callister 8th Ed.;

Diamagnetism

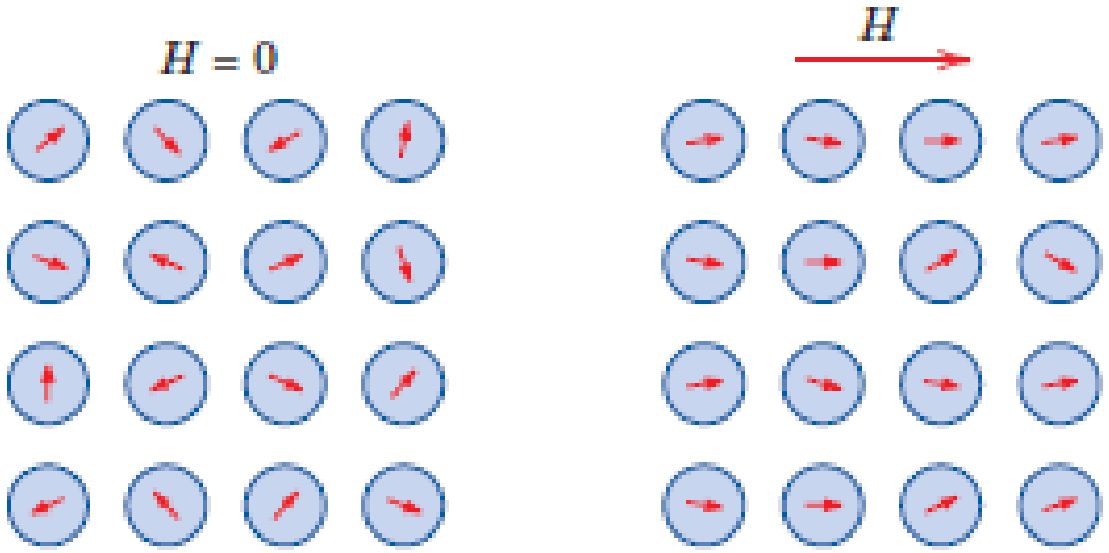


- Completely filled shells or subshells.
- Total cancellation of orbital and spin moments
- a very weak form of magnetism that is nonpermanent and persists only while an external field is being applied.
- It is induced by a change in the orbital motion of electrons due to an applied magnetic field because external magnetic field acting on the atoms slightly unbalances their orbiting electrons and creates small magnetic dipoles within the atoms, which oppose the applied field, and this action produces a negative magnetic effect to the applied field.

- The magnitude of the induced magnetic moment is extremely small, and in a direction opposite to that of the applied field.
- The relative permeability *r is less than unity (however, only very* slightly), and the magnetic susceptibility is negative; that is, the magnitude of the *B* field within a diamagnetic solid is less than that in a vacuum.

A weak, negative, repulsive reaction of a material to an applied magnetic field.

Paramagnetism



Both diamagnetic and paramagnetic materials are considered nonmagnetic because they exhibit magnetization only when in the presence of an external field.

- For some solid materials, each atom possesses a permanent dipole moment by virtue of incomplete cancellation of electron spin and/or orbital magnetic moments
- Permanent Dipoles. Randomly oriented when no field is present. Permanent dipoles align with an external field. No interaction between adjacent dipoles.
- In the absence of an external magnetic field, the orientations of these atomic magnetic moments are random
- as the dipoles align with the external field, they enhance it, giving rise to a relative permeability *r that is greater than* unity, and to a relatively small but positive magnetic susceptibility.

A weak, positive, attractive reaction of a material to an applied field

Diamagnetic and Paramagnetic materials

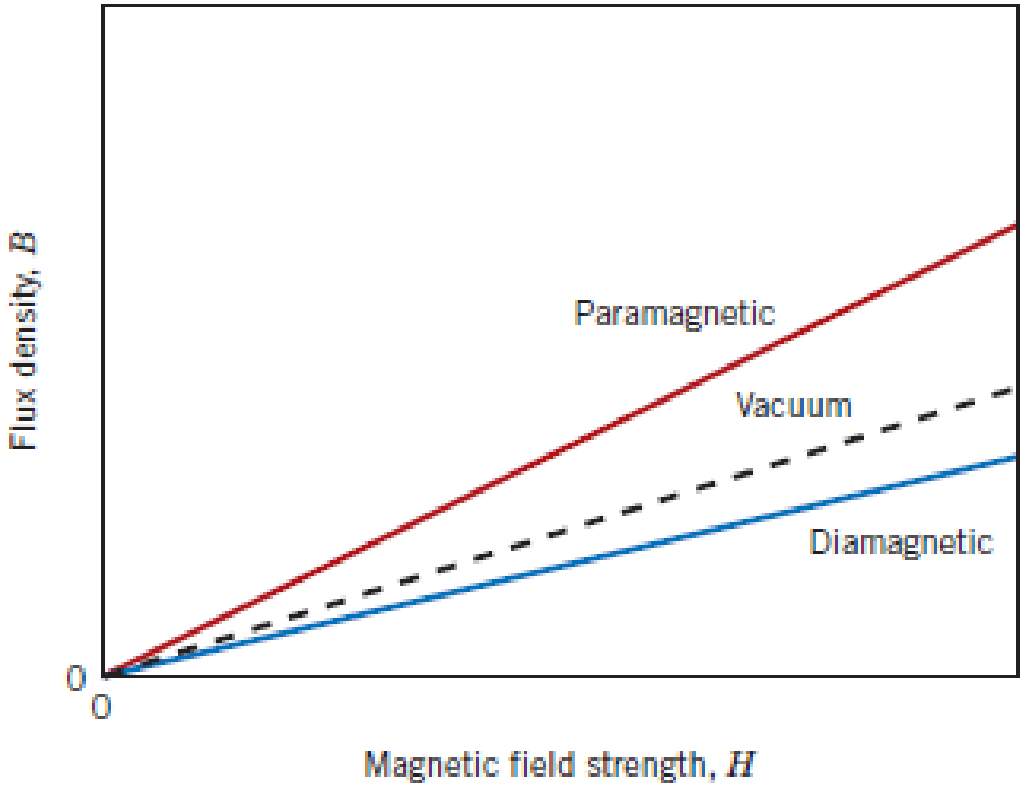
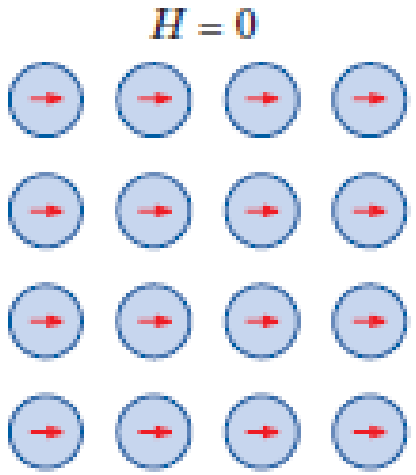


Table 20.2 Room-Temperature Magnetic Susceptibilities for Diamagnetic and Paramagnetic Materials

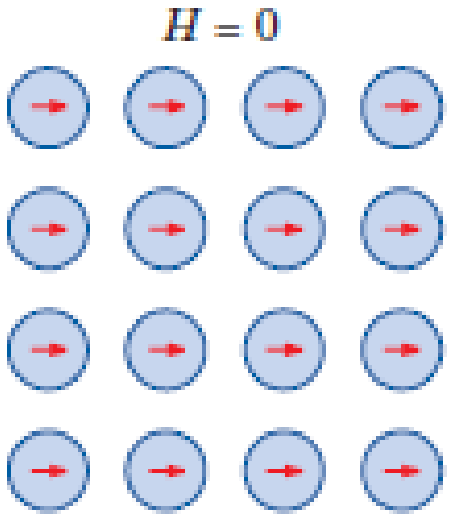
Diamagnetics		*Paramagnetics*	
Material	*Susceptibility χ_m (volume) (SI units)*	*Material*	*Susceptibility χ_m (volume) (SI units)*
Aluminum oxide	-1.81×10^{-5}	Aluminum	2.07×10^{-5}
Copper	-0.96×10^{-5}	Chromium	3.13×10^{-4}
Gold	-3.44×10^{-5}	Chromium chloride	1.51×10^{-3}
Mercury	-2.85×10^{-5}	Manganese sulfate	3.70×10^{-3}
Silicon	-0.41×10^{-5}	Molybdenum	1.19×10^{-4}
Silver	-2.38×10^{-5}	Sodium	8.48×10^{-6}
Sodium chloride	-1.41×10^{-5}	Titanium	1.81×10^{-4}
Zinc	-1.56×10^{-5}	Zirconium	1.09×10^{-4}

Ferromagnetism



- Certain metallic materials possess a permanent magnetic moment in the absence of an external field, and manifest very large and permanent magnetizations
- Permanent magnetic moments in ferromagnetic materials result from atomic magnetic moments due to *uncancelled electron* spins as a consequence of the electron structure
- There is also an *orbital magnetic moment* contribution that is small in comparison to the spin moment.
- In a ferromagnetic material, *coupling interactions of e- spin* cause net spin magnetic moments of adjacent atoms to align with one another, even in the absence of an external field.
- These are the characteristics of ferromagnetism, and they are displayed by the transition metals iron (as BCC -ferrite), cobalt, nickel, and some of the rare earth metals such as gadolinium (Gd).

Ferromagnetism



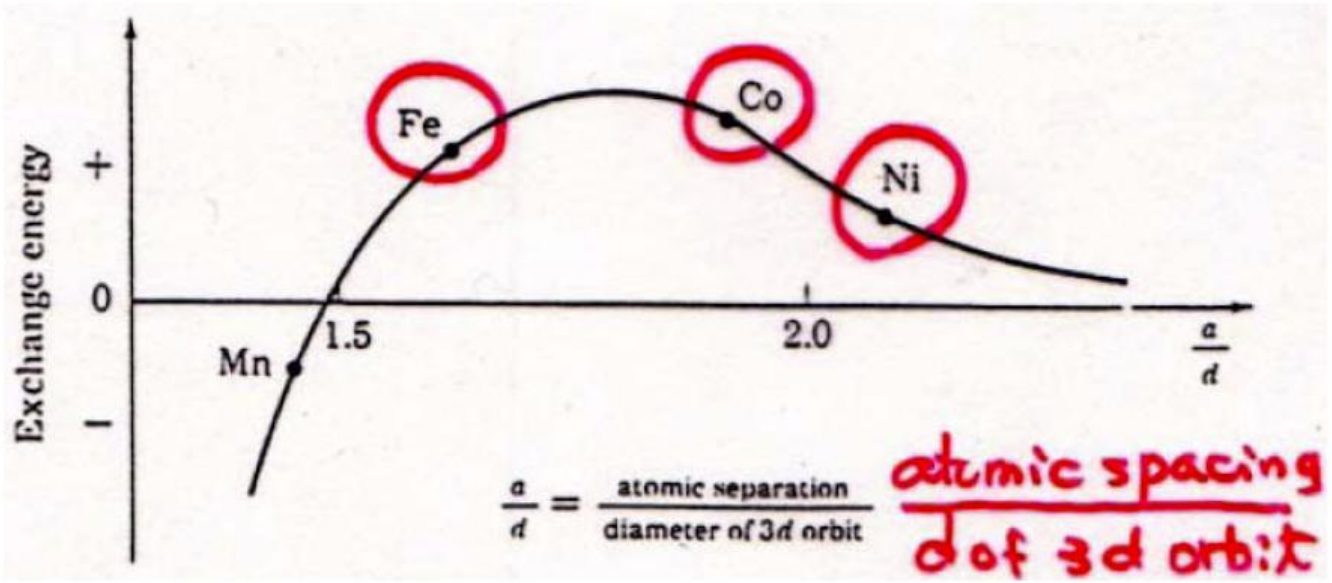
Fe, Co and Ni →ferromagnetic materials

Cr and Mn→not ferromagnetic materials, Why?

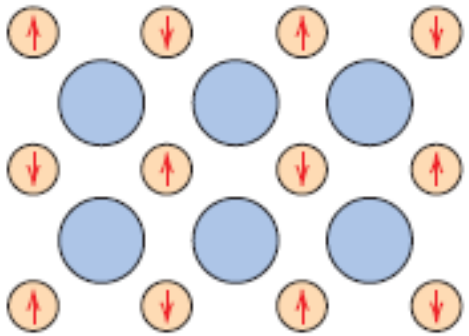
(they all have unpaired 3d electrons)

Magnetic exchange interaction energy → Energy associated with the coupling of individual magnetic dipoles into a single magnetic domain. Only when this exchange energy is **positive** → material be ferromagnetic.

Magnetic exchange energy is related to the *ratio of atomic spacing to 3d orbit*, must be in the range of 1.4 to 2.7.



Antiferromagnetism





- Magnetic moment coupling between adjacent atoms or ions occurs in materials which are not ferromagnetic
- If the coupling of electron spins results in antiparallel alignment then spins will cancel each other and no net magnetic moment will arise. This is known as antiferromagnetism
- Manganese oxide is **a ceramic oxide material** that **is ionic in character**, having both Mn2+ and O2- ions. No net magnetic moment is associated with the O2- ions (O2 ion is not magnetic), because there is a total cancellation of both spin and orbital moments.
- However, the Mn2+ ions possess a net magnetic moment that is predominantly of spin origin. These Mn2+ ions are arrayed in the crystal structure such that the moments of adjacent ions are antiparallel
- Obviously, the opposing magnetic moments cancel one another, and, as a consequence, the solid as a whole possesses no net magnetic moment.

Ferrimagnetism

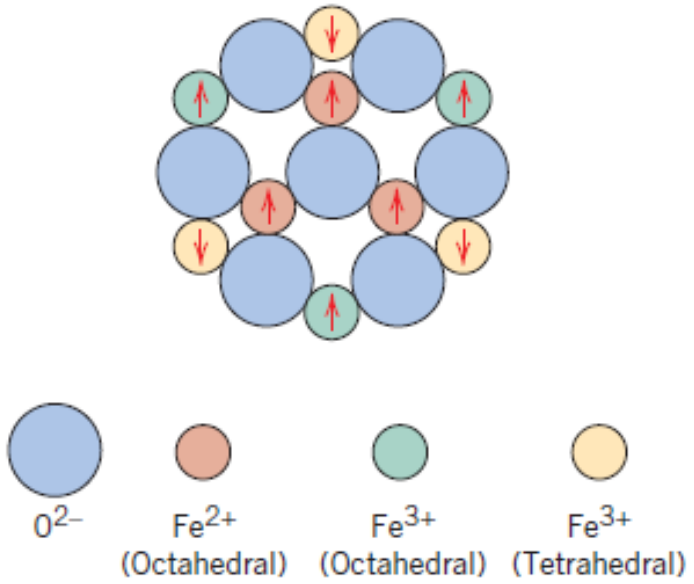


Table 20.3 The Distribution of Spin Magnetic Moments for Fe^{2+} and Fe^{3+} Ions in a Unit Cell Fe_3O_4[a]

Cation	*Octahedral Lattice Site*	*Tetrahedral Lattice Site*	*Net Magnetic Moment*
Fe^{3+}	↑ ↑ ↑ ↑ ↑ ↑ ↑ ↑	↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓	Complete cancellation
Fe^{2+}	↑ ↑ ↑ ↑ ↑ ↑ ↑ ↑	—	↑ ↑ ↑ ↑ ↑ ↑ ↑ ↑

- Some ceramics also exhibit a permanent magnetization, termed **ferrimagnetism**
- The macroscopic magnetic characteristics of ferromagnets and ferrimagnets are similar; the distinction lies in the source of the net magnetic moments
- Certain ionic solids having a general formula MFe2O4 where M is any metal show permanent magnetism , termed ferrimagnetism due to partial cancellation of spin moments.
- The antiparallel coupling between Fe3+ (half in A site and half in B site) moments cancels each others but Fe2+ moments are alighned in the same direction and result in net magnetic moment.

Antiferromagnetism vs Ferrimagnetism

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Antiferromagnets: 

Near-neighbor spins are antialigned. No hysteresis.
Examples: MF_2, MO (M = Mn, Fe, Co, Ni, Cu), IrMn …

Ferrimagnets: 

Near-neighbor spins are antialigned, and are also not compensated (the spins don't cancel). Hysteresis and magnetization like in a ferromagnet.
Examples: Fe_3O_4, $CoFe_2O_4$, $Y_3Fe_5O_{12}$, $BaFe_6O_{19}$ …

Magnetic Hysteresis

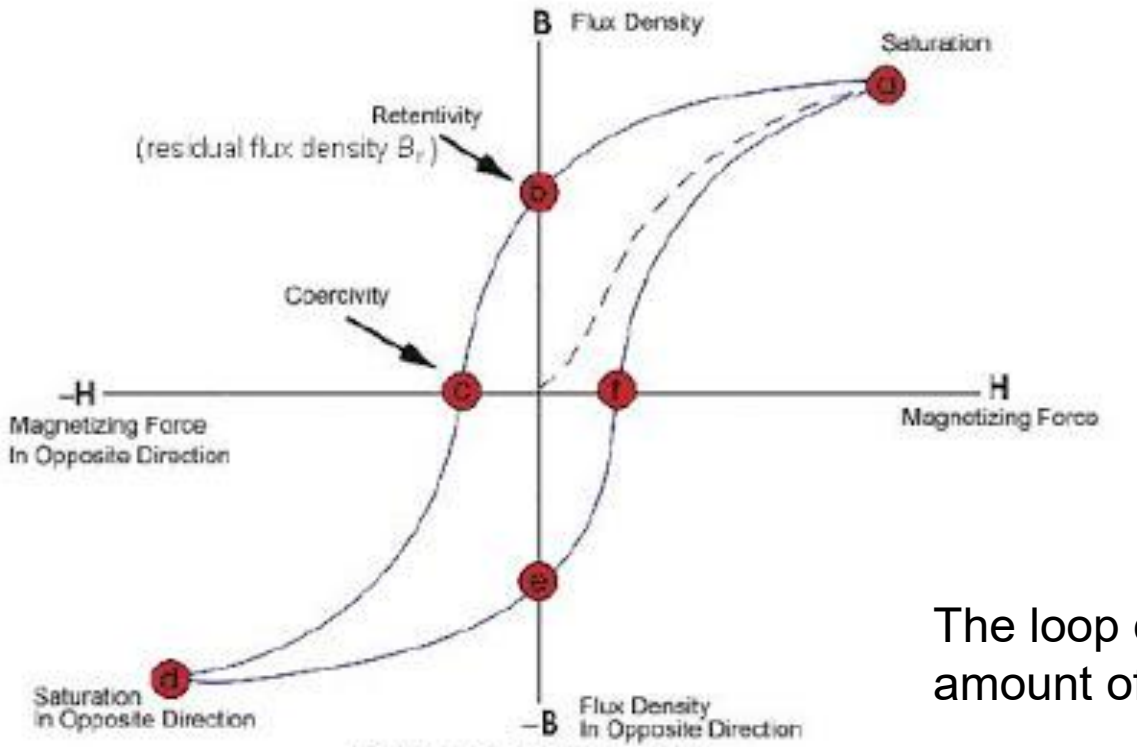


Figure 4.6: Hyteresis loop.

A magnetic hysteresis (hysteresis loop):

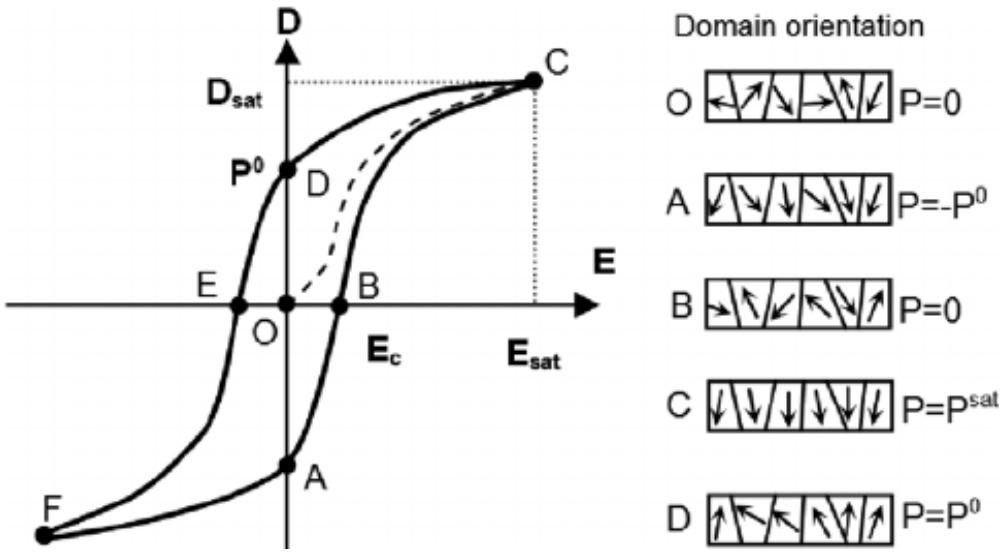
- magnetizing force (H) versus the magnetic flux density (B) of a ferromagnetic material.
- The curvature of the hysteresis is characteristic of the type of material being observed
- Can vary in size and shape (i.e. narrow or wide).

The loop can be generated by using a Hall Effect sensor to measure the amount of magnetic field at various points –

- when in the presence of a magnetic field,
- when it is removed from the magnetic field, and
- when a force is applied to bring the magnetic flux back to zero.

These loops are important in the memory capacity of devices for magnetic storage of data on computer disks.

Magnetic Hysteresis



What will happen if you reduce the magnetic field from saturation magnetization?

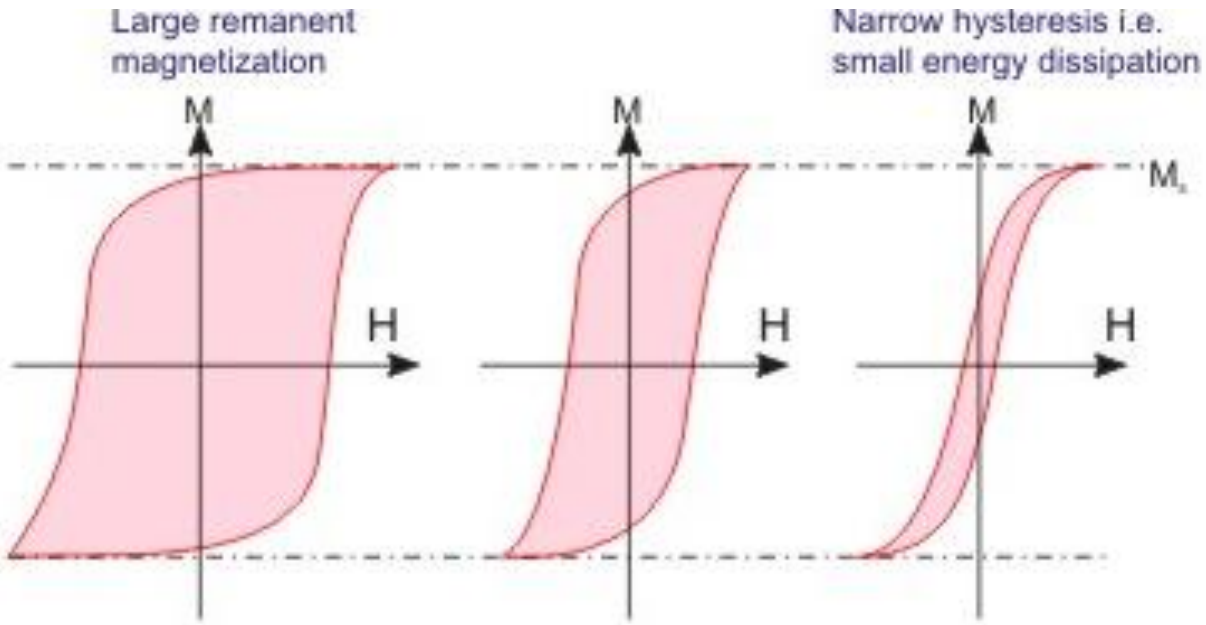
- If the field is reduced from the saturation by magnetic reversal a hysteresis look develops.
- As the field is reversed, the favorably oriented domains tend to align in the new direction.
- When H reaches zero at point P some of the domains still remain aligned in the previous direction giving rise to a residual magnetisation called Remanence,
- The reverse field strength (-Hc) at which magnetization is zero (E), is called coercivity.

The area within a loop represents a magnetic energy loss per unit volume of material per magnetization-demagnetization cycle in the form of heat

https://www.doitpoms.ac.uk/tlplib/ferromagnetic/hysteresis.php

Significance of Hysteresis loop

$$\text{Power Dissipated} = \text{Area of } B-H \text{ Curve} \times \text{Volume of Material} \times \text{Frequency}$$



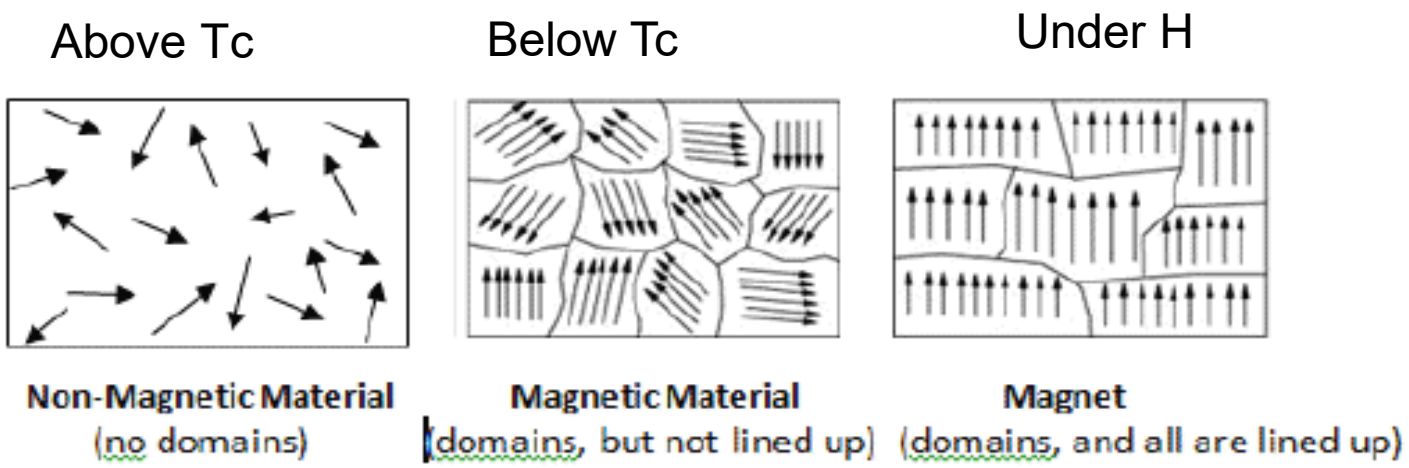
In soft magnets:

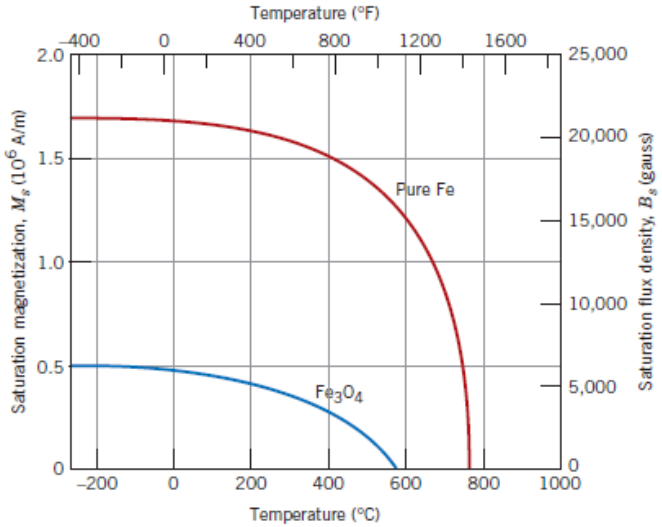
- Narrow hysteresis->>> low loss
- High permeability and low coercivity due to easy movement in domain wall
- Easy to magnetize and demagnetize

Another feature which is used to explain hard and soft magnets is the area of B-H or M-H curve which represents the power dissipated as heat and is expressed as So, since a hysteresis curve implies loss of a certain amount of energy, typically soft magnets show lower energy losses as compared to hard magnets. Soft magnets with reasonably large M_r and smaller H_c are useful.

https://nptel.ac.in/courses/113104005/lecture30a/30_5.htm

Temperature Effect on magnetic properties





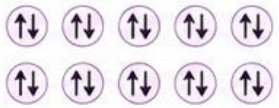
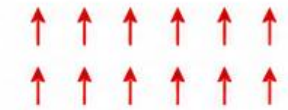
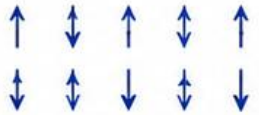
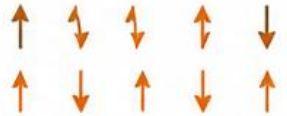
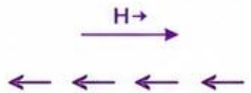
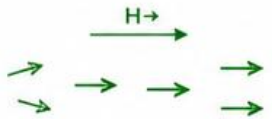
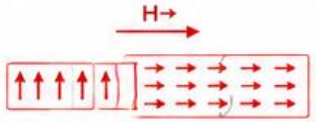
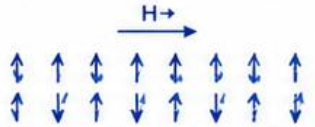
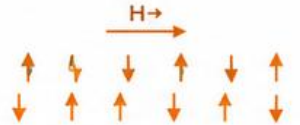
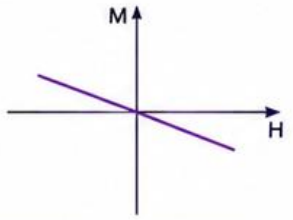
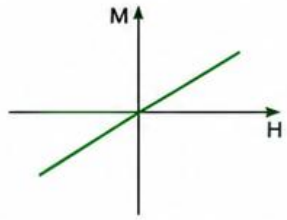
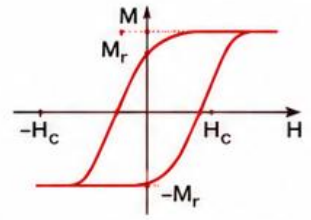
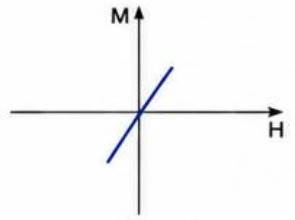
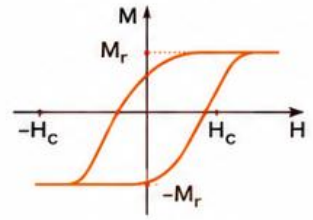
- Raising the temperature of a solid results in an increase in the magnitude of the *thermal vibrations* of atoms. The atomic magnetic moments are free to rotate; hence, with rising temperature, the increased thermal motion of the atoms tends to randomize the directions of any moments that may be aligned

- The saturation magnetization is a maximum at 0 K, at which temperature the thermal vibrations are a minimum. With increasing temperature, the saturation magnetization diminishes gradually and then abruptly drops to zero at what is called the ***Curie temperature, Tc*****.** The magnitude of the Curie temperature varies from material to material; for example, for iron, cobalt, nickel, and Fe3O4, the respective values are 768, 1120, 335, and 585C.

- Antiferromagnetism is also affected by temperature; this behavior vanishes at what is called the ***Néel temperature***. *At temperatures above this point, antiferromagnetic* materials also become paramagnetic

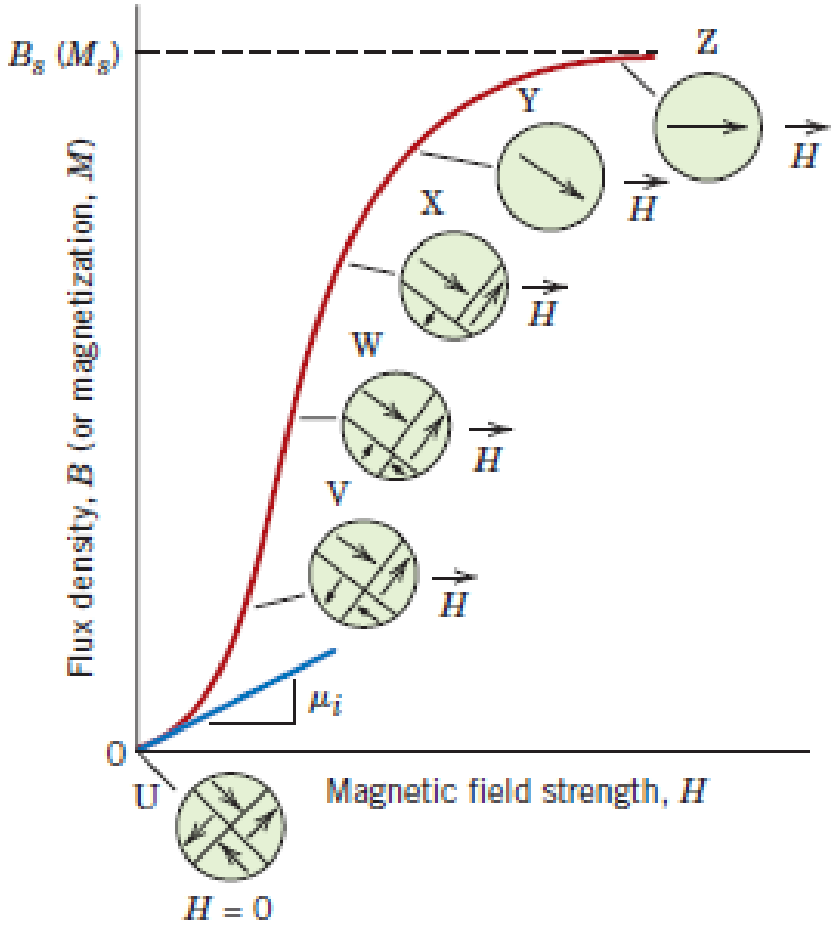
Five Basic Types of Magnetism

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	DIAMAGNETISM	PARAMAGNETISM	FERROMAGNETISM	ANTIFERROMAGNETISM	FERRIMAGNETISM
Atomic / Spin Arrangement (H = 0)	No permanent magnetic moment (all electrons paired) 	Permanent moments randomly oriented 	Spins align parallel (spontaneous magnetization) 	Neighboring spins align antiparallel and cancel 	Neighboring spins antiparallel but unequal → net magnetization 
Response to Applied Field (H →)	Induced moment opposite to applied field 	Moments tend to align with the field 	Domains grow and align with the field 	Slight canting of spins; small net moment 	Unequal moments lead to net alignment 
M–H Curve (at 300 K)					
Key Comments	• No permanent dipole moment • Induced moment opposite to H • Weak repulsion from magnets • Present in all materials	• Permanent moments exist • Random due to thermal motion • Weak attraction to magnets • No hysteresis	• Strong parallel exchange interaction • Spontaneous magnetization below Curie temperature (T_C) • Hysteresis with large M_r and H_c	• Strong antiparallel exchange • Net moment ≈ 0 (if equal) • Below Néel temperature (T_N) • No hysteresis (or very small)	• Antiparallel but unequal moments • Nonzero net magnetization • Hysteresis similar to ferromagnets • Below Curie temperature (T_C)
Examples	Bi, Cu, Ag, Au, Water, Graphite	Al, Pt, Mg, Cr, O_2, rare earth salts	Fe, Co, Ni, Gd, Dy, Sm	MnO, NiO, FeO, Cr_2O_3, MnF_2	Fe_3O_4 (Magnetite), $CoFe_2O_4$, $NiFe_2O_4$
Magnetic Susceptibility (χ)	$\chi < 0$ (very small, ~10^{-6} to 10^{-5})	$\chi > 0$ (small, ~10^{-6} to 10^{-3})	$\chi >> 1$ (very large, nonlinear)	$\chi \approx 0$ (very small) (positive above T_N, negative below T_N)	$\chi >> 1$ (large, nonlinear)

Magnetism arises from electron spin and orbital motion and their exchange interactions. Temperature can destroy magnetic order: Ferromagnetism and ferrimagnetism disappear above Curie temperature (T_C); Antiferromagnetism disappears above Néel temperature (T_N).

Domain boundary movement and Saturation Magnetisation



- When a magnetic field is applied to a ferromagnetic materials, domain tends to align (similar to a charge particle deforms after applying electric field) in the direction of the field by domain boundary movement and hence the flux density or magnetization increases .
- As the field strength increases domains which are favorably oriented to field direction grow at the expense of the small and un-oriented ones.
- All the domains are aligned to the field direction at high field strength and the material reaches at saturation magnetization Ms
- The initial slope of the curve at H=0 is called initial permeability μ which is a material property

https://www.doitpoms.ac.uk/tlplib/ferromagnetic/hysteresis.php

Ferromagnetism to superparamagnetism (transition to nanomagnetism)

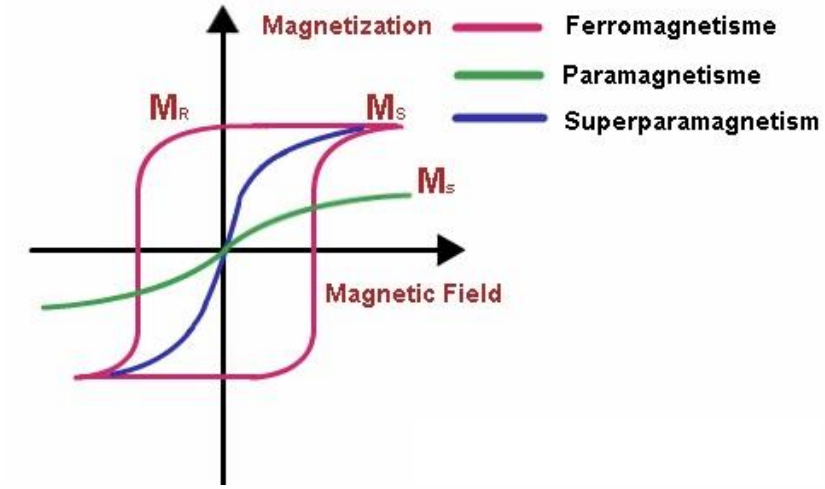
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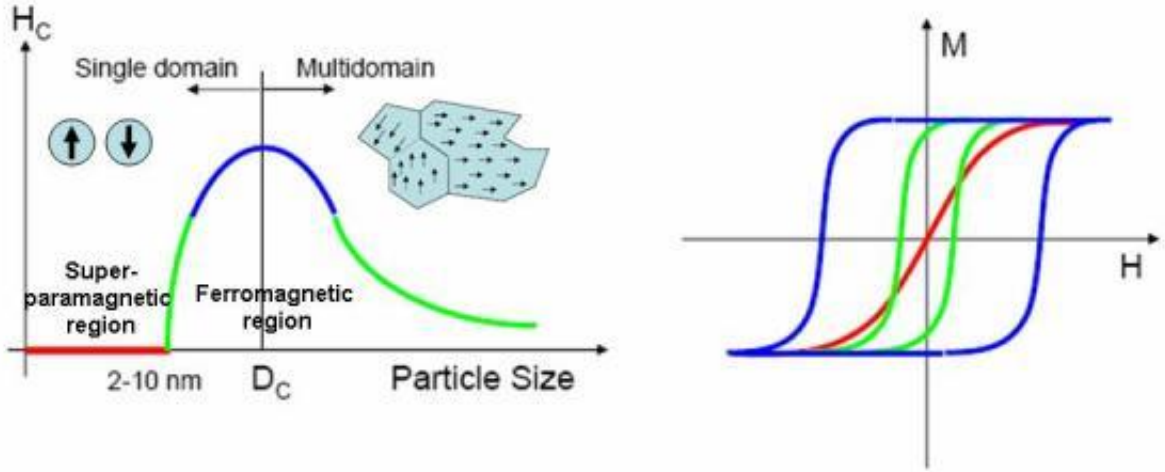
What is 'Super' about Superparamagnetism
Very fine ferromagnetic particles have very short relaxation times, even at room temperature, and behave superparamagnetic.

Superparmagnetic materials have:

- A high saturation magnetiszation (Ms)
- Zero coercivity (Hc) and
- Zero remanence (Mr),
- Single domain

Making it to be distinguished from ferromagnetism and paramagnetism - Superparamagnetism is a size effect of ferromagnetism.





The influence of magnetic particle size on magnetic properties. The coercivity changes with the particle size.

https://www.science20.com/mei/blog/blocking_temperatu

https://mappingignorance.org/2014/03/05/superparamagnetic-nanoparticles-and-the-separation-problem/

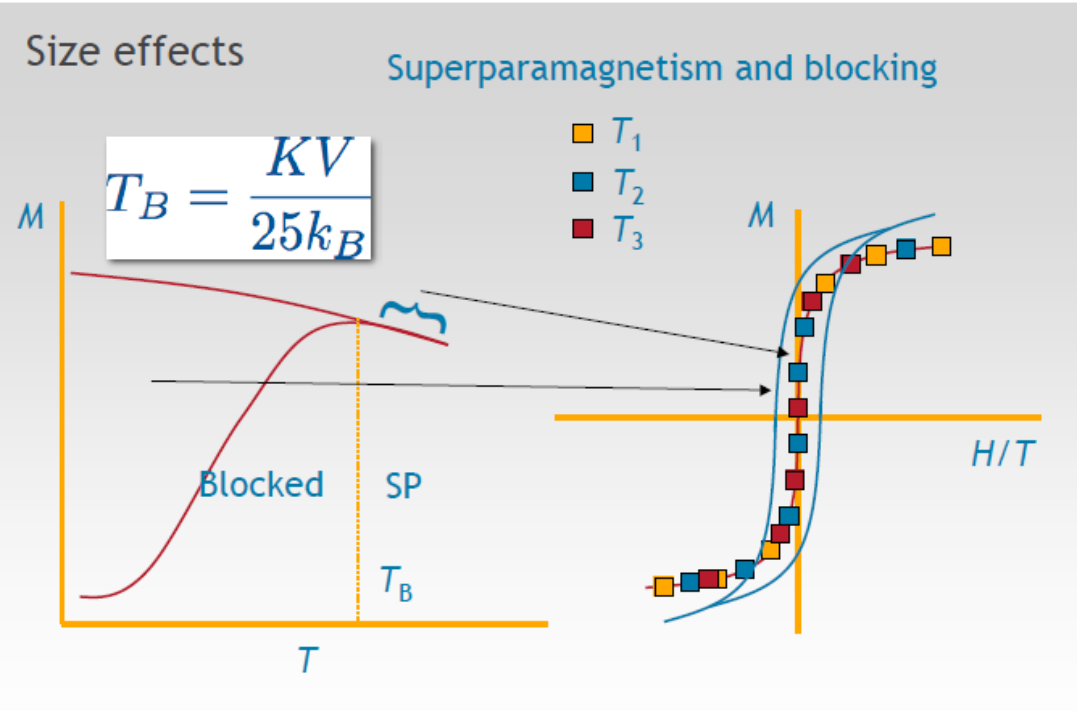
Ferromagnetism to superparamagnetism (transition to nanomagnetism)

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Some important features about Superparamagnetism

- Their behavior is paramagnetic (lower Ms) but their magnetization values are typical of ferromagnetic substances (higher Ms and zero Hc and Mr).

- The individual particles have normal ferromagnetic movements but very short relaxation times, enabling them to rapidly follow directional changes in an applied field.



- Magnetization curves measured at different temperatures superimpose when magnetization (*M*) is plotted as a function of field (*H*)/temperature (*T*).

- This also demonstrates that superparamagnetism can be destroyed by cooling.

- The temperature at which this occurs is called the blocking temperature (T_B) and is dependent linearly on volume and the magnitude of the crystal field anisotropy.

K is the effective magnetic anisotropy constant, V is the magnetic particle volume, kB is Boltzmann's constant, and TB is the **blocking temperature**.

Problem solving

You synthesize two batches of Fe_3O_4 nanoparticles:
Batch A: diameter DA=8 nm
Batch B: diameter DB=20 nm

Assume spherical particles, room temperature T=300K, Cu Kα XRD confirms single-crystal particles, and the effective anisotropy energy density is $K=1.0 \times 10^5$ J m^{-3}
Use the "blocking criterion" based on the Néel–Arrhenius law with a typical measurement window ($\ln(\tau m/\tau 0)\approx 25$), i.e.
Tb = KV/25kb
KV=25kbTb

Energy barrier, Eb = KV
Thermal fluctuation energy, Et = 25kbTb
Blocked/ferromagnetic if $EB=KV \gtrsim 25$ kBT, superparamagnetic if $EB \ll 25$ kBT.

Tasks
1. Compute the particle volumes VA and VB.
2. Compute the energy barriers EB,A and EB,B.
3. Compare each to 25kBT and decide whether each batch is superparamagnetic or blocked at 300 K.

Given Constants:
$kB=1.380649\times10^{-23}$ J/K (Boltzmann constant)
$25kBT=25\times kB\times 300\approx 1.035\times10^{-19}$ J
Sphere volume $V=4/3\pi r^3$

A biomedical company is developing **superparamagnetic iron oxide nanoparticles (SPIONs)** for magnetically targeted drug delivery. The nanoparticles should respond strongly to an external magnetic field so that they can be guided toward a target site. However, once the external magnetic field is removed, the particles should have negligible remanence and coercivity to minimise magnetic aggregation.

The nanoparticles have an effective magnetic anisotropy constant, $K=1.3\times104\ Jm-3$
and can be approximated as spherical, non-interacting, single-domain particles.

For the measurement timescale being considered, assume the blocking condition is
$KV=25kBTB$

where
$kB=1.38\times10-23\ JK-1$.

The nanoparticles will operate at body temperature:
$T=310\ K$.

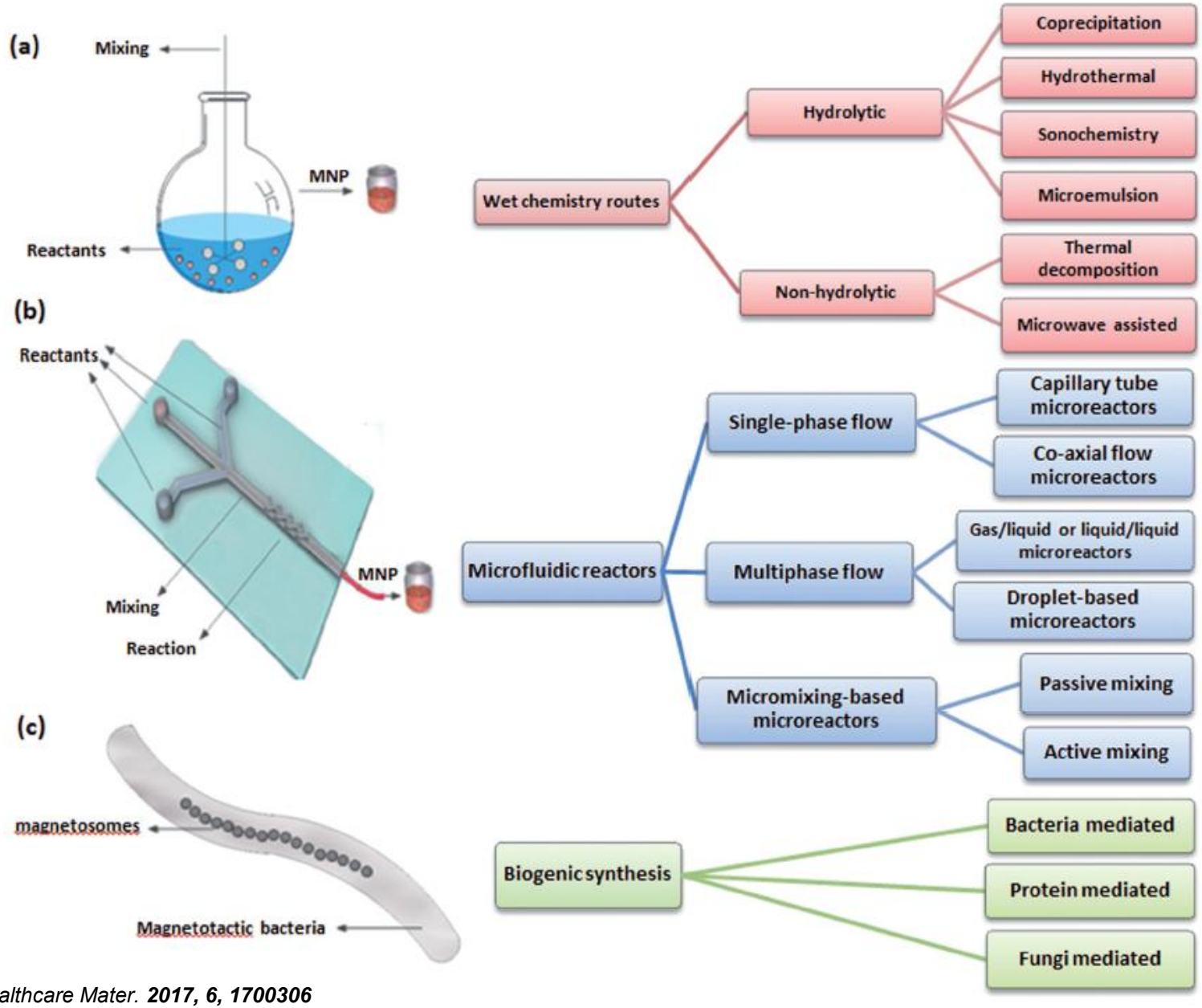
(a) Calculate the maximum approximate particle diameter corresponding to a blocking temperature of 310 K.

(b) The manufacturing team can produce particles with average diameters of 15 nm, 25 nm, and 35 nm. Which would you recommend for superparamagnetic operation at body temperature? Justify your answer using the energy-barrier concept.

(c) A colleague says:
“The 35 nm particles are better because larger particles contain more magnetic material and therefore will respond more strongly to the applied magnetic field.”

Do you agree? Explain the engineering trade-off between magnetic response and magnetic stability.

List all the synthesis techniques and comparison



Adv. Healthcare Mater. ***2017, 6, 1700306***

List all the synthesis techniques and comparison

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Synthesis route	Coprecipitation	Hydrothermal
Reaction temperature	20–90	180–220
Reaction period	Minutes	hours
Solvent	Water	Water, ethanol
Production	High	Medium
Advantages	Mild reaction conditions Synthesis in H_2O Easy surface modification Ease conversion to maghemite Ease scale up	Improved size control Narrow size distribution Synthesis in H_2O Tunable magnetic properties Applicability for continuous synthesis
Disadvantages	Broad size distribution Polydispersity Low reproducibility Uncontrolled oxidation Difficulty in adjusting the pH of the reaction	High temperature High pressure Need for special reactors or autoclaves

Adv. Healthcare Mater. ***2017, 6, 1700306***

List all the synthesis techniques and comparison

Synthesis route	Sonochemistry	Microemulsion
Reaction temperature	20–90	20–50
Reaction period	Minutes	hours
Solvent	Water	Water
Production	Medium	Low
Advantages	Accelerated reaction time Enhanced mixing by non-invasive source of energy Relatively narrow size distribution Good dispersivity High crystallinity	Improved size control Relatively narrow size distribution Ease of size tunability Uniform magnetic properties
Disadvantages	Need for ultrasound wave irradiation equipment	Low reaction yield Poor crystallinity Difficulty to remove surfactants Need for large amounts of organic solvents

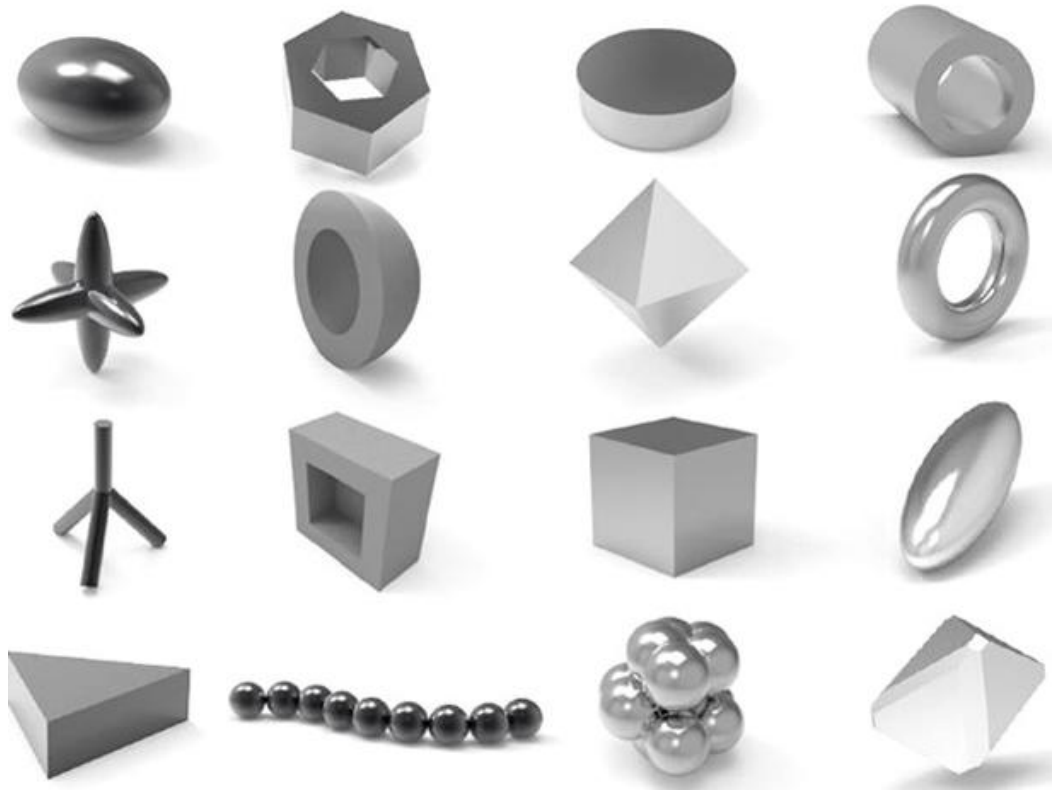
List all the synthesis techniques and comparison

Synthesis route	Thermal decomposition	Microwave assisted
Reaction temperature	100–320	160–210
Reaction period	hours	Minutes
Solvent	Organic compound	Organic compound
Production	High	High
Advantages	Narrow size distribution High crystallinity High size control Tunable magnetic properties	Narrow size distribution, high crystallinity, water dispersible, Rapid processing, high reaction rate, reduced reaction time, high yield of product along with simplicity and high energy efficiency
Disadvantages	Toxic organic solvents High temperature Need for phase transfer	Need for organic solvent Limited penetration depth of the microwave

Parameters influence magnetic properties:

Ms (Saturation magnetisation), Mr (Remanance), Hc (Coercivity), Tb (Blocking temperature)

- Shape
- Size
- Composition
- Synthesis processes
- Sintering temperature
- Induced Porosity
- Density
- Shell-core structure



https://doi.org/10.1016/j.addr.2018.12.008

Case study 1:

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Shape and Size effect on magnetic properties of nanocomposites

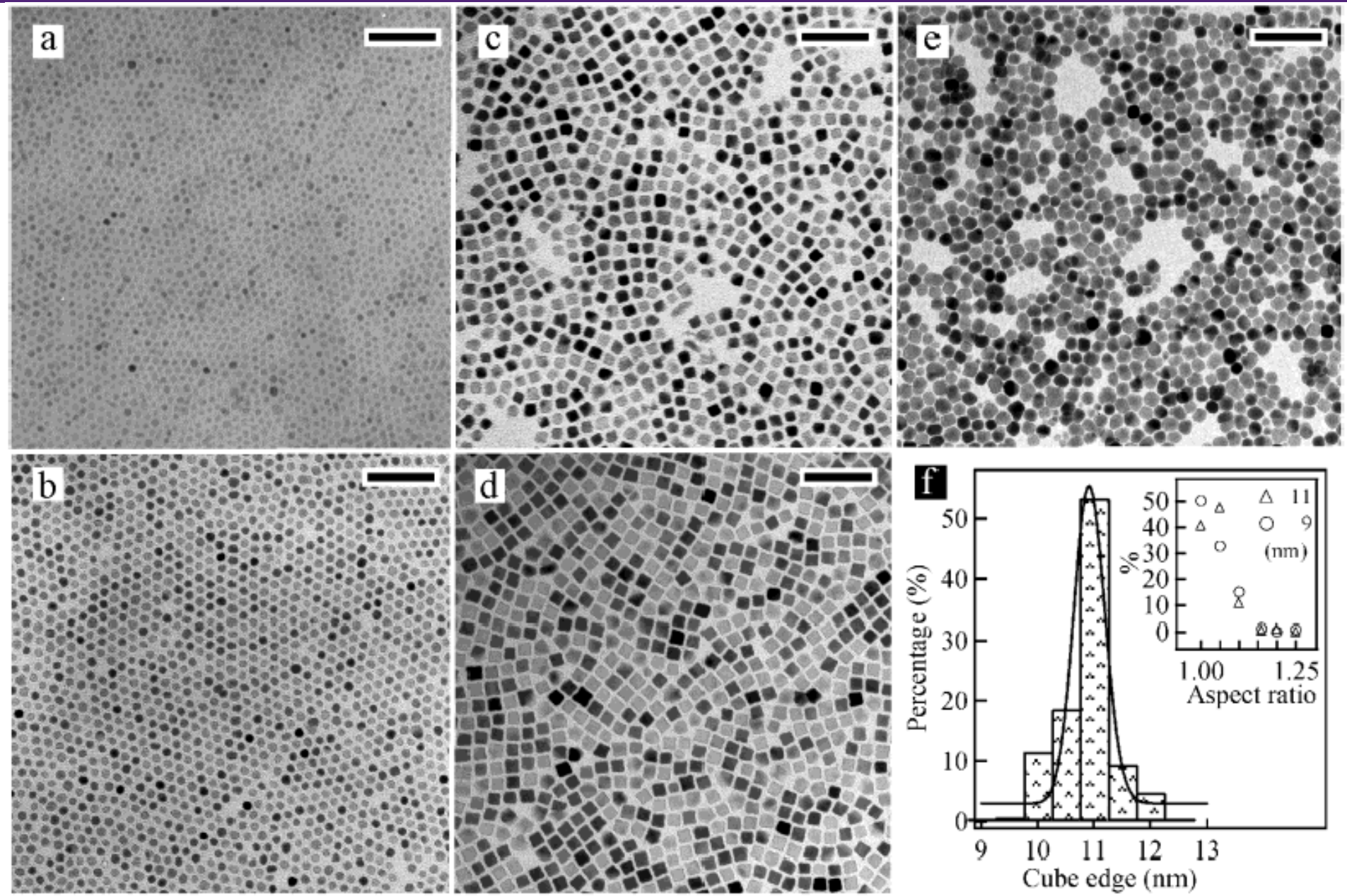
https://pubs-acs-org.ezproxy.library.uq.edu.au/doi/full/10.1021/ja049931r

Case study 1:

Monodisperse CoFe2O4 nanocrystals are synthesized by using a combination of a nonhydrolytic process and seed-mediated growth.

TEM micrographs with the scale bar as 50 nm. The size in panel (a) and (b) is 5.2 +/-1.1 and 7.9 +/-0.5 nm spherical CoFe2O4 nanocrystals, respectively. Panels (c) and (d) show 9.1 +/- 0.5 and 10.9 +/- 0.6 nm cubic CoFe2O4 nanocrystals, respectively. Panel (e) is 11.8 (1.3 nm spherical CoFe2O4 nanocrystals. Panel (f), as a representative histogram, shows the size distribution of cubic nanocrystals in panel (d) after measuring over 400 nanocrystals. The inset in panel (f) displays the aspect ratios of cubic nanocrystals in panels (c) and (d).

Growth reaction parameters such as heating rate, temperature, reaction time, ratio of seed to precursors, and ratio of oleic acid to oleylamine have been systematically studied for the control of size and shape of nanocrystals.



https://pubs-acs-org.ezproxy.library.uq.edu.au/doi/full/10.1021/ja049931r

Case study 1:

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- Precursors: coordination compounds of iron(III) and cobalt(II) acetylacetonate, $Fe(acac)_3$ and $Co(acac)_2$

- 5nm Seed preparation: 2 mmol of $Co(acac)_2$ + 40 mL of phenyl ether + 20 mmol of 1,2-hexadecanediol + 10 mL of oleic acid, + 10 mL of oleylamine --→ 140°C → droplet addition of 4 mmol $Fe(acac)_3$ + 20 mL of a phenyl ether solution → 260 °C → reflux for 30 min before being cooled down to room temperature-→ addition of ethanol and centrifuging-→spherical $CoFe_2O_4$ 5nm seed

- 8 nm spherical nanocrystals: 5 nm nanocrystals were used as seeds to grow 8 or 9 nm spherical nanocrystals in the solution of $Co(acac)_2$ and $Fe(acac)_3$ precursors. 100 mg of seeds was used in a particle growth solution consisting of 1 mmol of $Co(acac)_2$, 2 mmol of $Fe(acac)_3$, 10 mmol of 1-octadecanol, 5 mL of oleic acid, and 5 mL of oleylamine. Then, the solution temperature was raised to 260 °C at a rate of 10−15°/min and kept at reflux at 260 °C for 30 min. The nanocrystals precipitated out after ethanol was added.

Note: Monodisperse nanocrystals with a 9 nm diameter can be obtained using the same procedures but with increased concentration of precursors in the particle growth solution. Although nanocrystals with a larger diameter can also be prepared with 5 nm seeds, the quality of nanocrystals in term of size distribution usually deteriorated. monodisperse nanocrystals with larger diameters were typically produced by incrementally increasing the size of nanoparticulate seeds.

Growth reaction parameters such as heating rate, temperature, reaction time, ratio of seed to precursors, and ratio of oleic acid to oleylamine have been systematically studied for the control of size and shape of nanocrystals. The results indicated that heating rate and growth temperature control the shape of $CoFe_2O_4$ nanocrystals.

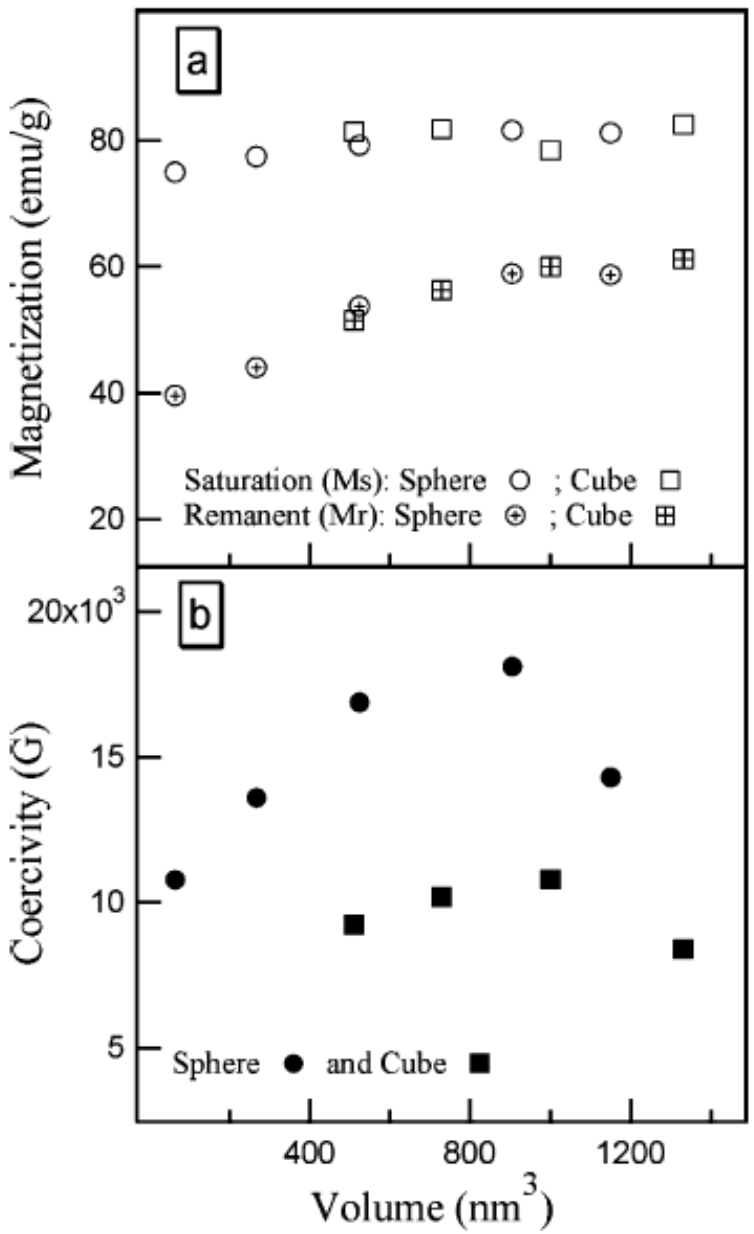
Due to the decisive effect of heating rate and growth temperature on shape control, Figure 1e shows almost spherical monodisperse $CoFe_2O_4$ nanocrystals with a 12 nm diameter, which were prepared by using nanocube seeds with an 8 nm edge length. Such a reversible shape formation clearly demonstrated that the shape of $CoFe_2O_4$ nanocrystals can be precisely controlled through the temperature and heating rate.

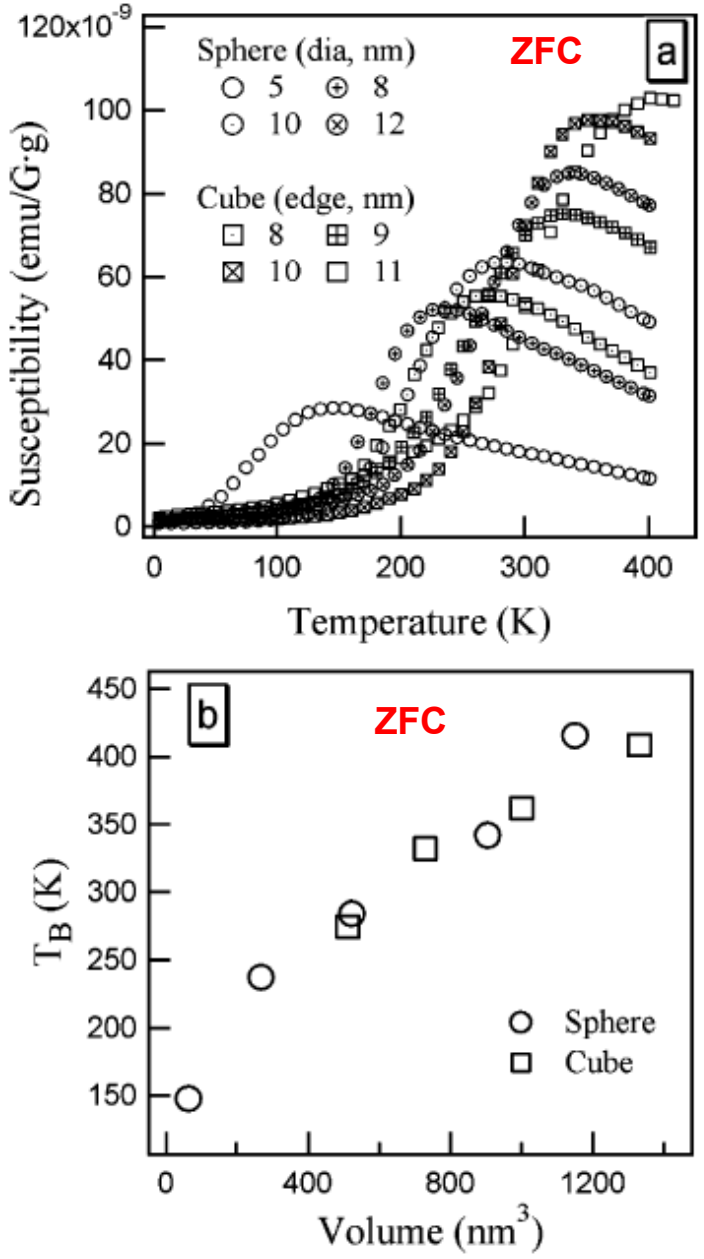
Size control → depends on synthesis parameter → The size of the nanocrystals was controlled through the ratio between the quantity of nanocrystalline seeds and the amount of precursors in the solution for seed-mediated growth.

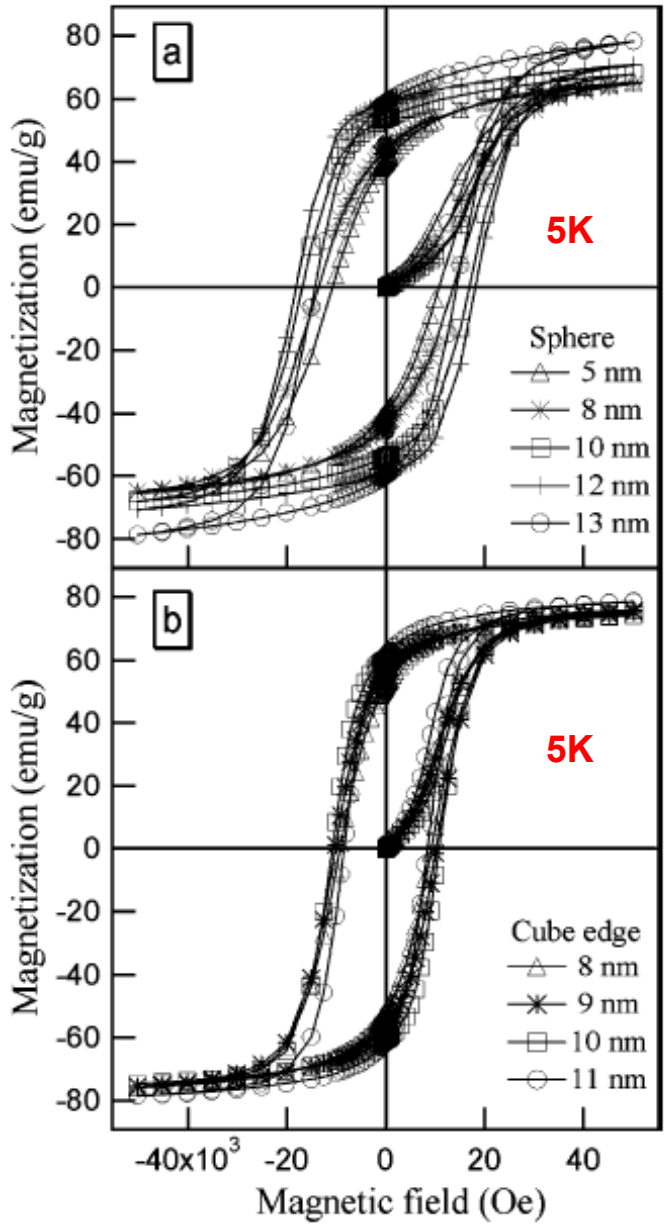
Shape control → depends on processing parameters → cubic $CoFe_2O_4$ nanocrystals were produced with an edge length of 8 or 9 nm when the temperature was raised only to 210 °C at a rate of 2−3°/min.

https://pubs-acs-org.ezproxy.library.uq.edu.au/doi/full/10.1021/ja049931r

Case study 1:


a
Magnetization (emu/g)
Saturation (Ms): Sphere ○ ; Cube □
Remanent (Mr): Sphere ⊕ ; Cube ⊞
b
20x10^3
Coercivity (G)
Sphere ● and Cube ■
Volume (nm^3)


120x10^-9
Susceptibility (emu/G·g)
Sphere (dia, nm)
5 8
10 12
Cube (edge, nm)
8 9
10 11
ZFC
a
Temperature (K)
b
ZFC
T_B (K)
Sphere
Cube
Volume (nm^3)


a
Magnetization (emu/g)
5K
Sphere
5 nm
8 nm
10 nm
12 nm
13 nm
b
5K
Cube edge
8 nm
9 nm
10 nm
11 nm
-40x10^3
Magnetic field (Oe)

Case study 2:

THE UNIVERSITY OF QUEENSLAND
AUSTRALIA

Effect of annealing temperature on magnetic phase transition in magnetic nanoparticles

https://www.sciencedirect.com/science/article/abs/pii/S0304885314012633

Case study 2



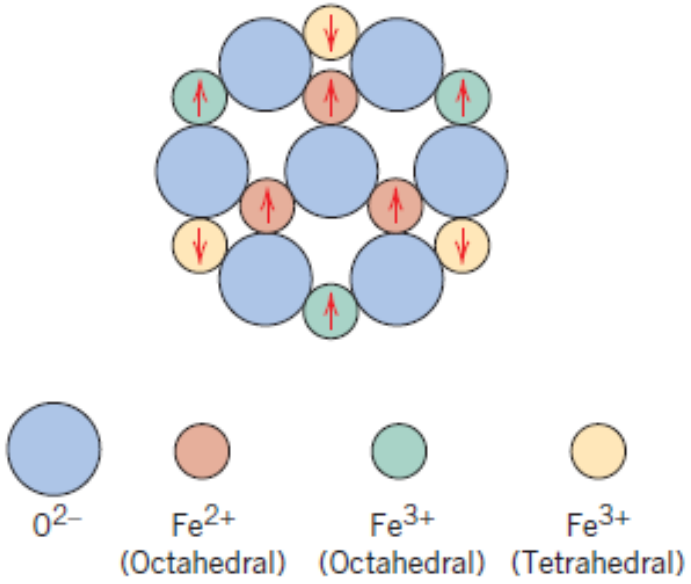


Table 20.3 **The Distribution of Spin Magnetic Moments for Fe^{2+} and Fe^{3+} Ions in a Unit Cell Fe_3O_4[a]**

Cation	*Octahedral Lattice Site*	*Tetrahedral Lattice Site*	*Net Magnetic Moment*
Fe^{3+}	↑↑↑↑ ↑↑↑↑	↓↓↓↓ ↓↓↓↓	Complete cancellation
Fe^{2+}	↑↑↑↑ ↑↑↑↑	—	↑↑↑↑ ↑↑↑↑

26	Fe	↑↓	↑↓ ↑ ↑ ↑ ↑		$[1s^2\,2s^2\,2p^6\,3s^2\,3p^6]\,4s^2\,3d^6$	$[Ar]\,4s^2\,3d^6$

Case study 2:

- Iron oxides, especially magnetite, are widely used in various fields, i.e, medical use in, for example, as contrast agents in MRI, targeting drug delivery, hyperthermia etc.

- Magnetite is thermodynamically unstable and can be oxidized to hematite, maghemite and other type of iron oxides in the presence of oxygen and depending on temperature conditions.

- This feature causes limitations in its applications under different temperature conditions.

- The investigation of the Fe_3O_4 NPs behavior at various temperatures or under different oxidizing conditions is necessary

- Magnetic properties are affected by phase transition processes

- Factors influencing the phase transitions:

Particle size, reaction medium (atmosphere, vacuum or inert gas such as Ar, N2, H2), preparation method, annealing time and heating rate in annealing.

Case study 2:

Synthesis Process:

$FeCl_2 \cdot 4H_2O$ and $FeCl_3 \cdot 6H_2O$ → 1:2 molar ratio dissolve in 25 ml DI water → Fe^{+2} and Fe^{+3}

Dropwise adding with magnetic stirring

200 ml of 1.5 mol/L NaOH (precipitator agent)

magnetic separation

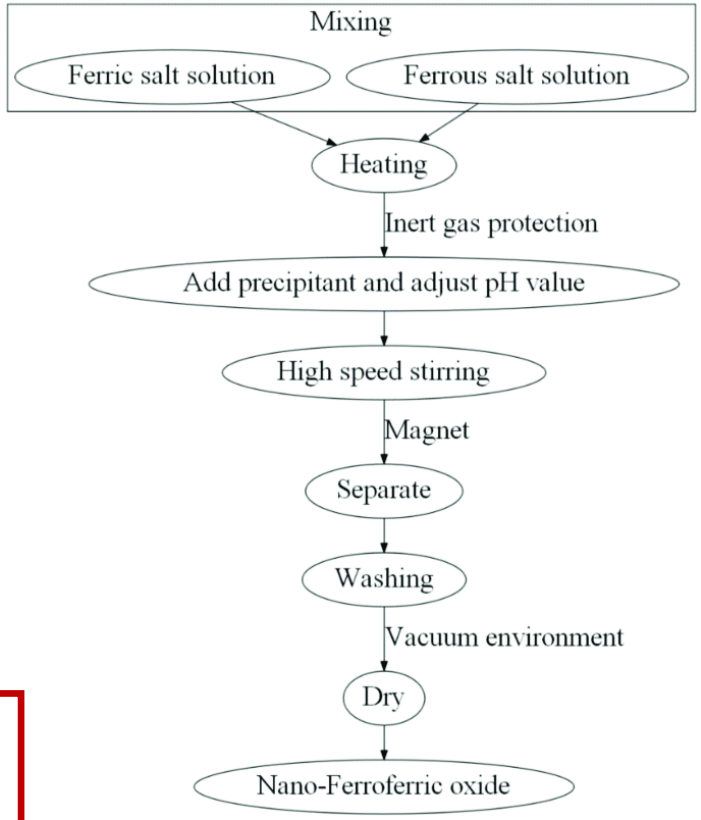
3 times wash with DI and ethanol

Black precipitations

Dry oven

50 C/24hrs

Fe3O4



$$Fe^{+2}(aq) + 2Fe^{+3}(aq) + 8OH^- \rightarrow Fe_3O_4 \downarrow + 4H_2O$$

annealing time (1 h) with heating rate of 5 °C/min

Phase transformation:

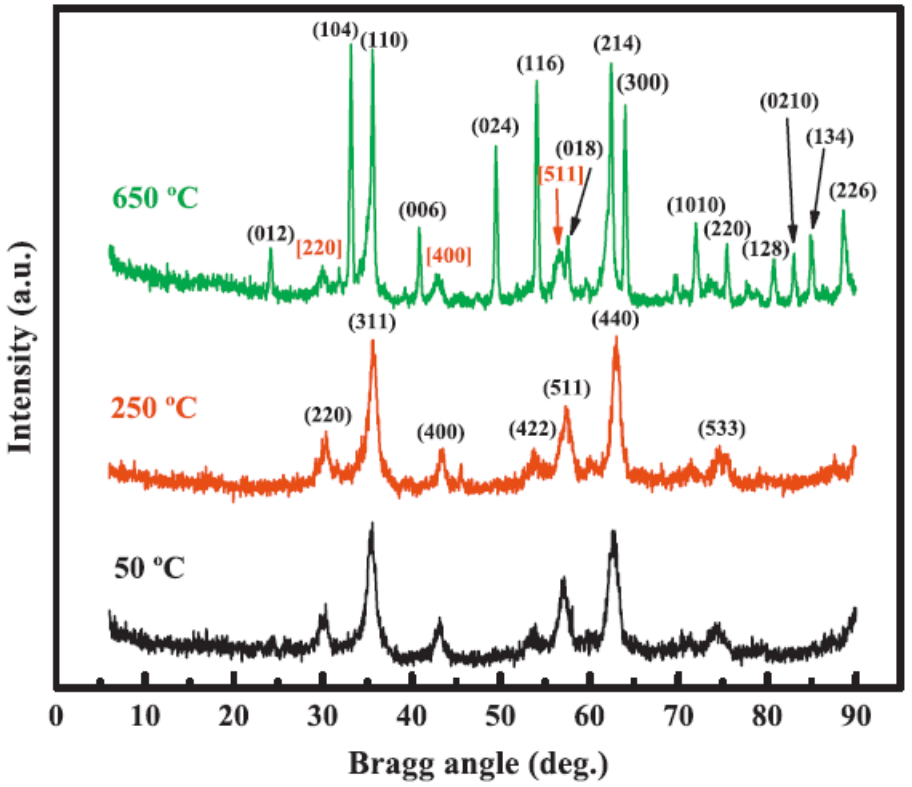
Fe3O4 —(250 C, 1 hr)→ Y-Fe2O3 —(550 C-650 C 1 hr)→ α- Fe2O3 —(750 C-800 C 1hr)→ α- Fe2O3

Fe3O4	Y-Fe2O3	α- Fe2O3	α- Fe2O3
Magnetite (Soft magnet)	Maghemite (Soft magnet)	hematite (Soft magnet)	hematite (hard magnet)
Superparamagnetic	Superparamagnetic	Superparamagnetic	Ferromagnetic

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DOI: 10.3390/pr6040033

Case study 2:



Summary of Debye–Scherrer size of the samples.

Annealing temperature (°C)	Debye–Scherrer size (nm)
50	6.6
250	7.5
350	8.8
450	9.8
550	12.7
650	23.9
750	31.2
850	37.6

No difference between the diffraction peaks of Fe3O4 NPs annealed at 50 (magnetite) and 250 °C (maghemite)

Difficult to distinguish magnetite from maghemite except precise calculation of d-value (lattice spacing) of (311) peak is varied from 2.531 to 2.518 Å (Raman spectroscopy)

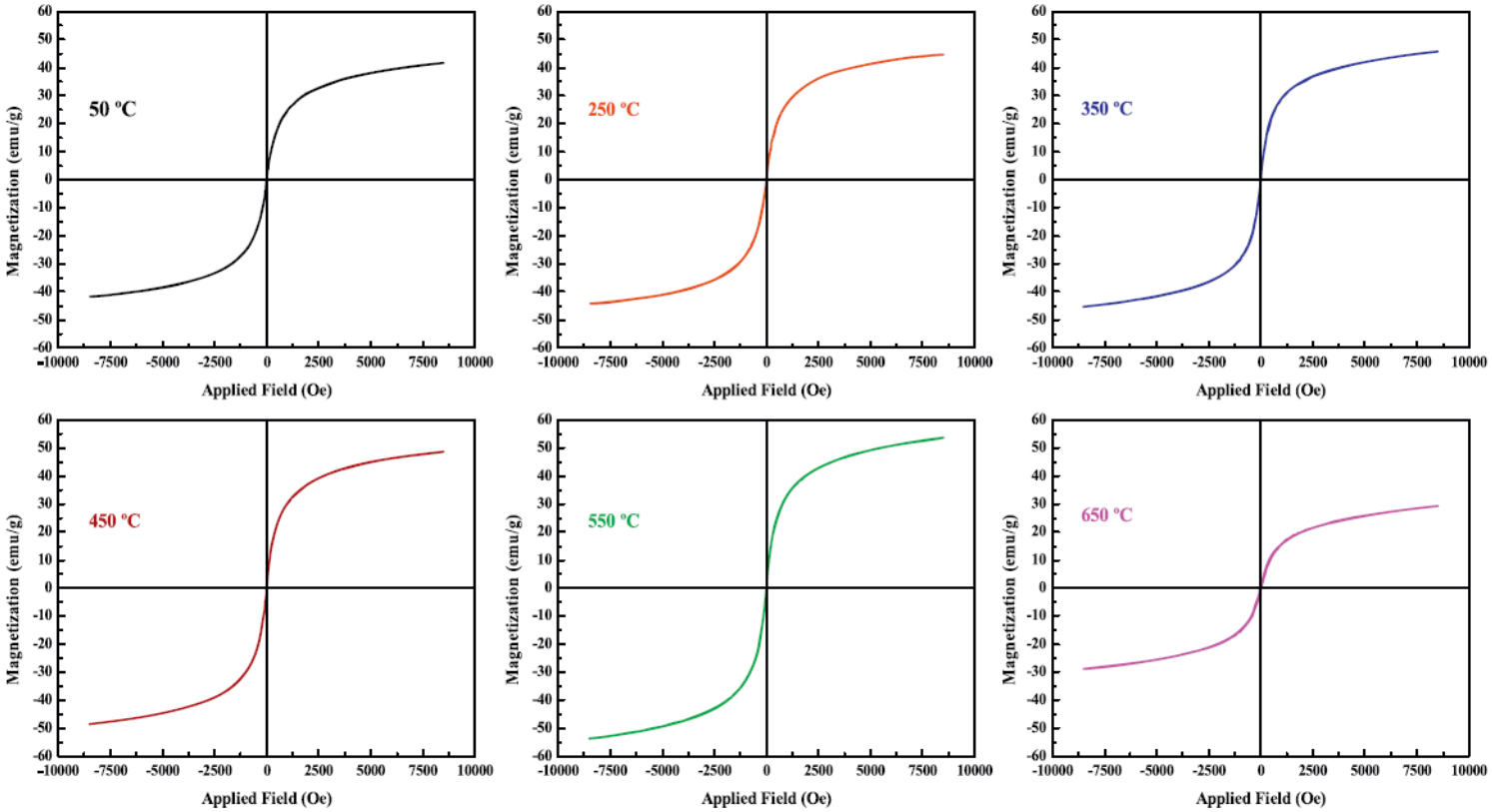
The magnetite (Fe3O4)-to-hematite (α-Fe2O3) conversion started at 650 °C

The average crystallite size of the prepared Fe3O4 NPs after annealing at different temperatures was calculated by the Debye– Scherrer equation:

$$D = \frac{k\lambda}{\beta \cos\theta}$$

Crystallite size of the NPs increased on increasing the annealing temperature due to the grain growth which is attributed to the coalescence of the particles by solid-state diffusion

Case study 2:



Room-temperature hysteresis loops of as-synthesized Fe3O4 NPs after annealing at 50, 250, 350, 450, 550, 650, 750 and 850 °C.

Langevin equation

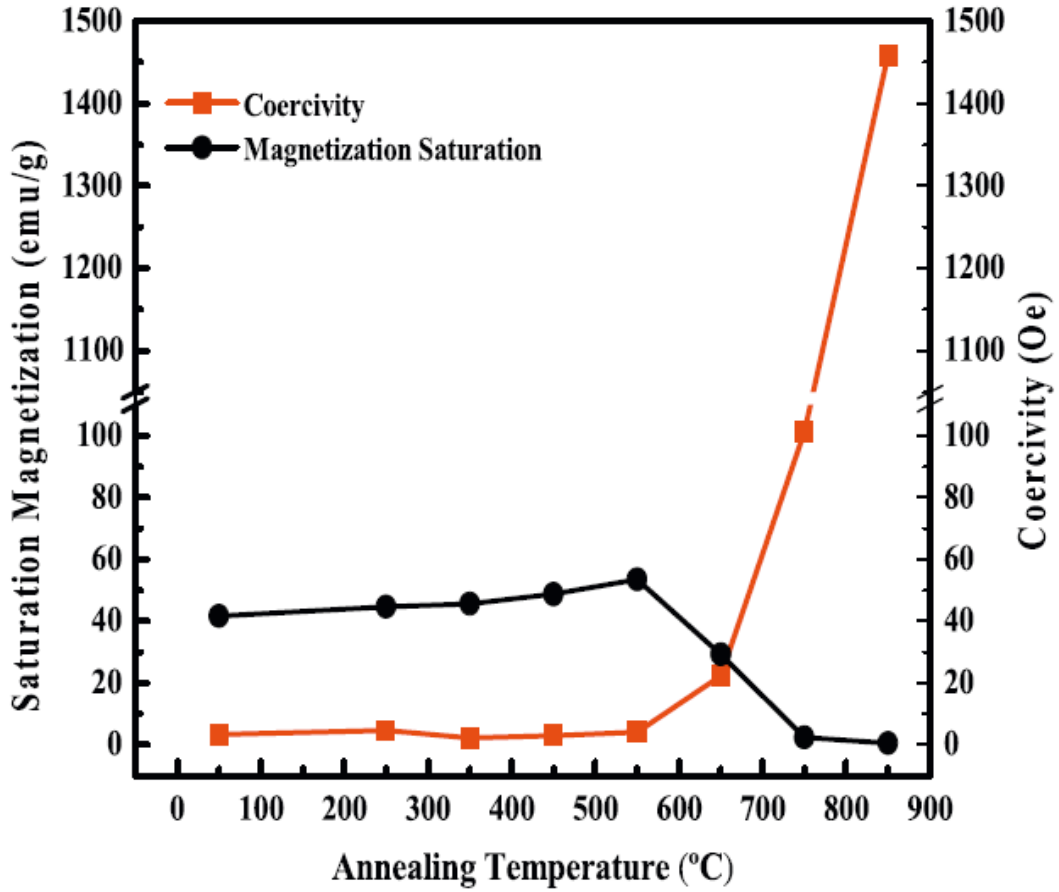
$$M = M_s \left[\coth\left(\frac{\mu B}{k_B T}\right) - \left(\frac{k_B T}{\mu B}\right) \right]$$

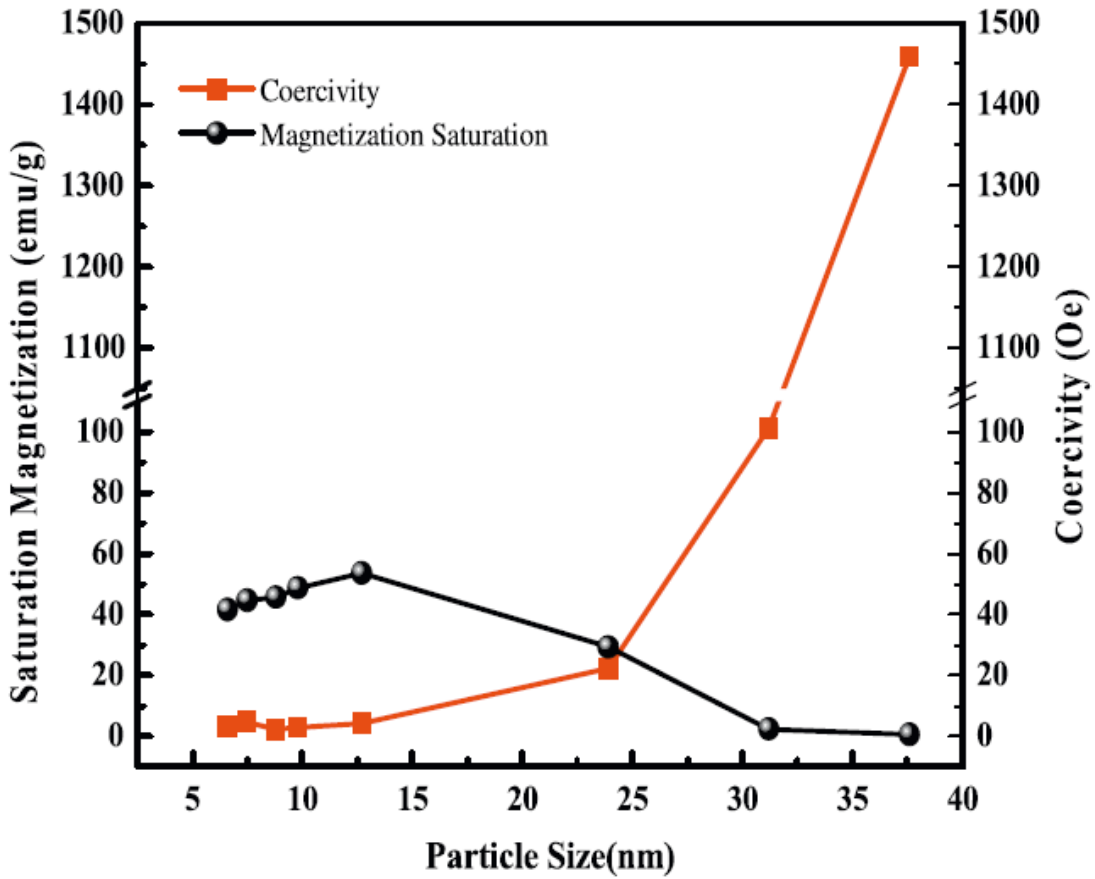
$$M_s = \mu N$$

μ is the magnetic moment of each particle, T the absolute temperature, kB the Boltzmann constant, and Ms the saturation magnetization of the sample, N is the number of particles/weight of sample).

Case study 2:

Variation of saturation magnetization and coercivity with annealing temperature and particle size.





Can you recall last week's lecture about the hysteresis loop?

Transformation from superparamagnetic to ferromagnetic!!!

$$H_c = \frac{0.96K}{M_s}$$

The relationship between Hc, Ms and K at a certain temperatures [46]

Case study 2: Summary

- Fe_3O_4 (magnetite) nanoparticles(NPs) were synthesized using a co-precipitation method.
- Annealed at various temperatures between 50 and 850 °C for 1h in air (heating rate 5°C/min).
- The NPs were characterized by X-ray diffraction (XRD), Fourier transform infrared (FT-IR) spectroscopy and vibrating sample magnetometer (VSM).
- Fe_3O_4 NPs were converted to γ-Fe_2O_3 (maghemite) by annealing at 250 °C for 1h (XRD and FTIR)
- Then to α-Fe_2O_3 (hematite) on annealing in the range of 550–650 °C
- The average crystallite size of the NPs estimated by the Debye–Scherrer equation increased from 6.6 to 37.6nm by increasing annealing temperature from 50 to 850 °C
- The magnetite NPs were superparamagnetic (VSM)
- Converted to the maghemite with super- paramagnetic phase by annealing up to 550 °C
- A phase transition from soft (maghemite) to hard (hematite) ferromagnetic was occurred at annealing temperature 650 and 850 °C, respectively.
- The VSM analysis confirmed the XRD and FTIR results. Magnetite and maghemite have extreme similarity in crystal structure, and XRD is not sensitive enough to differentiate between them, FTIR needed.
- The saturation magnetization(Ms) of Fe_3O_4 NPs was increased from 41.69 to 53.61 emu/g by increasing annealing temperature from 50 to 550 °C, and then decreased intensively to 0.49emu/g after annealing at 850 °C.
- The coercive force (Hc) was reached to 4.20 Oe after annealing at 550 °C and then intensively increased to 1459.02 Oe for any further increasing of particle size up to 850 °C.

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Summary

- Understand the facts fundamental and basic terminology of magnetism
- Understand the origin of magnetism from atomic model
- Determination the magnetism in materials from their electronic configuration
- Understand the types of magnetism and their mechanism to distinguish
- Understand the effect of temperatures on magnetism
- Understand the magnetic hysteresis and their significance in practical application
- Understand the domain structure transformation (multi to single) and magnetic saturation from hysteresis loop
- Understand the ferromagnetism to superparamagnetism transformation (nanomagnetism)
- Understand the effect of size and shape effect on magnetic properties
- Understand the effect of annealing temperatures on magnetic properties of nanoparticles and mechanism of their phase transformation (magnetite to hematite)

Synthesis of magnetic nanoparticles for week 7's lab prac

The University Of Queensland
AUSTRALIA

1. Nanoparticle synthesis and their biofunctionalisation

2. Ni ferrite nanoparticle using hydrothermal process

3. All the equipment required such as incubator, hydrothermal chamber, vacuum oven, calcination, autoclave etc

4. Characterisation: SEM/magnetic properties